Public Comment on CMS-0042-NC: Health Technology Ecosystem RFI

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Introduction

I am Steve Yaskin, Founder and CEO of beHuman Corp., a healthcare technology company that delivers AI-powered preventive care to underserved populations through digital automation and payer-aligned reimbursement. Prior to founding beHuman, I was the founder and CEO of Health Gorilla, one of the first designated Qualified Health Information Networks (QHINs™) under the Trusted Exchange Framework and Common Agreement (TEFCA™). My leadership at Health Gorilla helped shape the foundation of the national health information exchange infrastructure.

beHuman builds on this infrastructure, offering a vertically integrated platform that leverages TEFCA to connect with clinical, payer, and lab data, enabling timely screening, longitudinal patient engagement, and precision care coordination. We aim to demonstrate how QHIN-based architectures can power scalable, consumer-facing care delivery.

In this public comment, I provide structured responses aligned with the RFI section order: Patients and Caregivers (PC), Providers (PR), Payers (PA), Technology Developers (TD), and Value-Based Care (VB). I am confident in providing feedback across all stakeholder categories due to my extensive experience in designing, implementing, and scaling health data interoperability solutions in collaboration with every sector of the healthcare ecosystem.

I advocate for mandatory TEFCA participation, universal digital identity through CSPs, and open access to interoperable health data as foundational pillars for the health technology ecosystem.

PATIENTS AND CAREGIVERS (PC)

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

The most impactful health management or care navigation apps would offer real-time, longitudinal access to clinical, behavioral, and benefit data in a single interface, integrated via TEFCA and powered by artificial intelligence (AI). These tools should extend beyond passive record access to provide contextual, proactive guidance, including information on screenings

due, follow-up appointments required, the impact of recent results on care, and available benefits or resources.

An ideal app would function as a "virtual care navigator," providing a unified experience across payers, providers, and healthcare systems. It would:

- Automatically pull in labs, diagnoses, medications, and past procedures
- Surface personalized health recommendations based on guideline adherence
- Send reminders for preventative screenings or chronic condition follow-ups
- Verify insurance coverage and out-of-pocket costs for care options
- Enable direct scheduling and digital intake
- Allow proxy access for caregivers managing multiple loved ones

a. Top things I (or caregivers) would like to do through digital health products include:

- Access all health records (labs, imaging, notes, medications) in one app
- Track preventative care gaps and chronic condition milestones
- Get alerts for abnormal labs or missed follow-ups
- Schedule appointments and receive reminders across multiple systems
- Understand which services are covered by insurance and the cost estimate
- Connect to local community resources (transportation, nutrition, housing)
- Manage care for dependents or elderly relatives through a shared dashboard

b. If I had a personal assistant to support my health needs, I would ask for help with:

- Summarizing my medical history across systems and surfacing key trends
- Identifying and scheduling necessary screenings or lab tests
- Reviewing benefits and helping me avoid unnecessary out-of-pocket costs
- Monitoring for changes in care plans or eligibility (e.g., MA plan switches)
- Translating clinical terminology into understandable language
- Coordinating follow-up with specialists and sending medical records
- Recommending low-cost or in-network options for care
- Tracking medications, renewals, and interactions
- Keeping a personalized health journal with recommendations and to-dos

These capabilities could—and should—be supported by intelligent, privacy-protected software solutions shortly. With TEFCA, CSP-backed digital identity, and AI-driven UX, such tools are not only possible but essential for democratizing access and empowering patients and caregivers to manage their health effectively.

PC-2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

Currently, most patients—including myself and my loved ones—do not have seamless access to a consolidated, longitudinal view of all health information in a single portal. Instead, we're forced to piece together fragmented data from disparate systems: EHR-specific portals,

pharmacy apps, insurance platforms, lab portals, and occasional PDF printouts. This fragmentation severely impairs care coordination, patient engagement, and proactive management of chronic conditions.

a. If so, what are some examples of benefits it has provided?

In rare cases where partial aggregation is possible, such as through Apple Health or Health Gorilla's Patient360, the benefits are substantial:

- Faster decision-making in emergencies when medication and diagnosis histories are readily accessible.
- Avoidance of duplicate testing, especially for labs and imaging.
- Improved care transitions between primary and specialist providers.
- More informed conversations with physicians, when the whole record is available to the patient and caregiver.

b. If not, in what contexts or for what workflows would it be most valuable to use one portal or system to access all such health information?

A single, unified portal would be most valuable in the following contexts:

- Managing multiple chronic conditions (e.g., diabetes, cancer, hypertension), especially when multiple specialists are involved.
- Coordinating care for elderly parents or children, where caregiver proxy access is essential.
- Navigating care transitions (e.g., hospital discharge to primary care follow-up)
- Participating in value-based or preventative care programs, which rely on visibility into gaps in care and patient adherence.
- Engaging in shared decision-making, where complete visibility into labs, imaging, and prior recommendations is necessary.

c. Were there particular data types, such as X-rays or specific test results, that were unavailable? What are the obstacles to accessing your own or your loved ones' complete health information electronically?

Yes, several critical data types are routinely missing from current systems:

- Radiology images and complete imaging reports
- Pathology and genetic test results
- Behavioral health and social determinants of health (SDOH) data
- Historical referrals, visit notes, and care plans
- Eligibility and real-time benefit information

Obstacles include:

- Lack of TEFCA participation by specialty providers, labs, and payers

- EHRs that don't expose FHIR APIs or limit patient-facing functionality
- Fragmented identity systems that prevent patients from linking data across organizations
- Low digital literacy or language barriers for caregivers and underserved populations
- User-unfriendly interfaces that bury critical insights in hard-to-navigate portals

Recommendation:

CMS should mandate TEFCA participation for all CMS-contracted payers and certified Electronic Health Record (EHR) vendors, enforce digital identity credentialing (e.g., Login.gov, ID.me) to enable secure access and proxy relationships, and require third-party applications to consolidate clinical and claims data in user-friendly formats. Only with a unified infrastructure can we provide patients and caregivers with the tools they need to manage their health conditions and navigate care with confidence effectively.

PC-3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

Yes, several health management, care navigation, and personal health record apps could be beneficial to Medicare beneficiaries and their caregivers, especially if certified under CMS's TEFCA Individual Access Services (IAS) framework.

TEFCA IAS-certified apps that integrate longitudinal health data from across QHIN-connected networks can offer comprehensive, patient-authorized access to clinical, claims, and social determinants data. These tools would allow beneficiaries to:

- View all of their medications, lab results, referrals, and care plans in one place.
- Manage appointments and transitions of care with real-time ADT alerts.
- Track eligibility and coverage details directly from payers.
- Access educational resources personalized to their conditions.

Examples of promising applications include:

- Health Gorilla's Patient360, which aggregates clinical records across QHINs, could be expanded to include patient-facing navigation functions.
- Apple Health and Google's Health Connect, which could be enhanced with TEFCA IAS certification, could be used to retrieve trusted health records from participating networks.
- Medicare Blue Button-enabled apps that provide claims visibility, which, if integrated with TEFCA data, would offer a more holistic patient experience.

By certifying these tools under TEFCA IAS, CMS can signal trustworthiness, ensure adherence to privacy and security standards, and create an ecosystem of interoperable, equitable, and accessible apps for all Medicare beneficiaries and their caregivers.

PC-4. What features are missing from apps you use or that you are aware of today?

Most health-related apps today fall short in offering integrated, actionable, and context-aware features. The most significant missing capabilities include:

- Access to complete, longitudinal health records, including labs, imaging, notes, and benefit data
- Real-time insurance eligibility and cost transparency
- Smart care navigation that surfaces next steps based on clinical history and guideline adherence
- Contextual clinical decision support (CDS) for patients and caregivers, not just providers
- Seamless scheduling and follow-up coordination across systems
- Support for proxies/caregivers to manage multiple dependents
- Language, accessibility, and offline features for underserved populations

Despite significant advancements in API capabilities (e.g., FHIR, SMART on FHIR), these functionalities remain fragmented across separate portals or app categories—often siloed by EHR vendors, laboratories, or payers.

a. What apps should exist but do not yet? Why do you believe they do not exist yet?

Apps that should exist but don't at scale include:

- A TEFCA-enabled care navigation app that aggregates clinical, benefit, and cost data for the patient, recommends the following actions, and allows real-time scheduling
- A caregiver dashboard that allows authorized users to manage health for children, aging parents, or spouses with shared records and alerts
- A benefits-integrated screening manager that recommends age-appropriate cancer or chronic condition screenings and confirms coverage in real time
- A digital front door that unifies pre-visit intake, insurance verification, history review, and post-visit follow-up across providers and payers

Why these don't exist:

- Lack of mandatory TEFCA participation by labs, payers, and specialty providers
- No universal digital identity credentialing to link records across systems securely
- Fragmented APIs and inconsistent FHIR implementation across vendors
- High development burden due to proprietary interfaces and a lack of standardized workflows
- Limited CMS policy support for patient-facing digital health infrastructure

b. What set of workflows do you believe CMS is uniquely positioned to offer?

CMS is uniquely positioned to support workflows that are cross-payer, cross-provider, and standards-enforced. These include:

- Longitudinal care gap analysis across FFS, MA, and Medicaid for all beneficiaries
- Screening and vaccination tracking tools, aligned with USPSTF and ACIP guidelines
- Real-time ADT event notifications and transitions of care summaries
- Eligibility-aware provider directories, showing in-network options by plan
- Automated authorization status and benefits validation APIs
- Care coordination dashboards for ACOs, Community Health Centers, and MSSP participants
- Equity dashboards that track disparities in engagement, outcomes, and SDOH data

By enabling these workflows through standardized APIs and TEFCA-backed data access, CMS can catalyze the next generation of patient- and caregiver-facing tools, bridging gaps that commercial developers cannot solve in isolation.

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

To increase adoption of digital health products, CMS and its partners must focus on three key pillars: usability, trust, and value clarity. Patients and caregivers will only adopt digital tools if they are easy to use, provide clear benefits, and are backed by credible standards that protect privacy and ensure data accuracy.

CMS should:

- Certify and promote digital health tools that meet national standards for usability and interoperability.
- Mandate accessibility features, such as multilingual support, large text, offline use, and proxy access.
- Tie adoption to real value, such as reduced out-of-pocket costs, faster appointments, or proactive reminders for life-saving care.
- Publicly score apps on accessibility, language support, digital equity features, and TEFCA/QHIN integration.

a. What role, if any, should CMS have in reviewing or approving digital health products based on efficacy, quality, or impact? What criteria should be used?

CMS should not act as a traditional FDA-style gatekeeper but rather play a meaningful role as a signaler of quality and trust, especially in contexts where data interoperability and equity are crucial.

CMS could maintain a “Trusted CMS-Connected” registry of digital health apps that:

- Integrate with TEFCA/QHINs or certified FHIR APIs
- Use CSP-backed digital identity for secure patient and proxy access
- Demonstrate usability through accessibility audits and real-world patient testing
- Support low-bandwidth and multilingual use for underserved populations
- Provide clear data provenance and transparency around sources

- Demonstrate positive impact on adherence, engagement, or outcomes through peer-reviewed or claims-linked evidence

CMS could also offer incentives to developers, such as access to testing environments or priority support, if their apps meet these criteria.

Technology/policy solutions to drive adoption:

- FHIR-based access to CMS claims + real-time benefit data for third-party apps
- Reimbursement incentives tied to digital engagement metrics (e.g., app-based follow-through on cancer screenings or chronic care visits)
- CSP credential bundling during Medicare enrollment to simplify secure access
- Targeted patient outreach campaigns in multiple languages, promoted via provider portals, MA plans, and community health centers

b. What changes would enable timely access to high-quality CMS and provider-generated data on patients?

CMS should:

- Mandate TEFCA participation for all Medicare-contracted providers and plans
- Publish FHIR APIs that expose Blue Button claims data, prior auth status, and eligibility in real time.
- Enable bulk and streaming access through trusted, patient-authorized apps and QHINs
- Require EHR vendors to support data submission and retrieval in a patient-centric, app-based workflow, not just provider portals.
- Establish a national identity + consent service, allowing patients to authorize data flows once and have that authorization persist across care settings.

With these changes, digital health products would become more useful, timely, and trusted, supporting better patient decisions, caregiver coordination, and ultimately, improved health outcomes.

PC-6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

To serve Medicare beneficiaries, many of whom are older, managing chronic conditions, or have limited digital literacy, digital health products must prioritize simplicity, clarity, and cultural sensitivity. The most important features include:

1. Large, high-contrast text and intuitive navigation

- Interfaces must be readable without zooming.
- Buttons and menus should follow predictable patterns, with minimal nesting of menus and clear, concise language.

2. Multilingual support

- Interfaces and instructions must be available in multiple languages, especially Spanish, Chinese, and other regionally relevant languages.
- Language preference should be set once and persist across apps and sessions

3. Offline or low-bandwidth functionality

- Apps should work in areas with poor connectivity, offering offline appointment reminders or summaries, and syncing data when reconnected.
- Support for SMS-based reminders and voice guidance should be built in

4. Voice and touch interaction options

- Voice-activated features (especially for visually impaired or dexterity-challenged users) can reduce friction
- Simplified, single-tap actions (e.g., "Tap to schedule" or "Tap to confirm medication") avoid confusion

5. Proxy access and shared dashboards

- Caregivers must be able to manage health records and appointments for family members with CSP-backed digital identity and documented consent
- Dashboards should indicate whose information is being viewed and what actions are pending

6. Accessibility compliance (WCAG 2.1 AA minimum)

- Apps should be tested for screen reader compatibility, closed captioning, and alternate input devices
- Compliance should be required for CMS digital certification and publicly scored

7. Simple onboarding and credentialing

- Patients should not have to remember multiple usernames or passwords
- CMS should promote federated digital identity (e.g., Login.gov, ID.me) and allow biometric or code-based login flows

8. Educational content in plain language

- Content should explain the "why" behind actions (e.g., why a screening matters)
- Visual summaries and short video guides can reduce drop-off and build confidence

Accessibility must be integrated into the certification process for any CMS-endorsed digital health product, rather than being treated as a secondary consideration. Products that cater to the

needs of digitally underserved populations will have the most significant impact on improving equity, engagement, and clinical outcomes in the Medicare population.

PC-7. If CMS were to collect real-world data on digital health products on health outcomes and related costs once they are released into the market, what would be the most effective means of doing so?

The most effective and scalable way for CMS to collect real-world data (RWD) on digital health product performance is to leverage TEFCA infrastructure, FHIR-based APIs, and CMS claims datasets to link product usage with measurable outcomes.

Key components of an ideal framework:

1. TEFCA-enabled usage monitoring

- Require QHINs and participating apps to log anonymized, de-identified usage events (e.g., appointment scheduled, medication reminder confirmed, care gap closed).
- These events should be timestamped and linked to digital identity tokens, not identifiable personal data.

2. Link to CMS claims and encounter data

- Use CMS beneficiary IDs and API tokens (with appropriate privacy protections) to connect app usage to outcomes such as:
    - Reduced hospitalizations or ED visits
    - Completed preventive screenings
    - Medication adherence
    - Cost-of-care metrics for specific episodes (e.g., post-op recovery)

3. Establish a CMS RWD clearinghouse or dashboard

- Aggregate app performance data across Medicare Advantage, FFS, and ACO models
- Allow developers to opt in for voluntary benchmarking and visibility (similar to app stores with quality scoring)
- Publicly showcase tools that demonstrate a positive return on investment (ROI), quality improvement, or equity impact.

4. Use standardized outcome definitions and APIs

- Define success metrics by use case (e.g., closed HEDIS gaps, reduction in avoidable admissions)
- Require digital health tools to support FHIR-based feedback endpoints that send structured usage and engagement data to CMS or approved intermediaries.

5. Incorporate patient-reported outcomes (PROs)

- Apps should optionally submit PROs (collected via short surveys or check-ins) that measure satisfaction, understanding, or changes in behavior.
- CMS could standardize the PRO format and reward apps that consistently capture these insights.

CMS is uniquely positioned to lead a national, claims-linked evaluation framework for digital health tools, modeled on real-world evidence standards used in pharmaceuticals. By leveraging TEFCA, FHIR, CSPs, and its own cost and quality data, CMS can validate the value of digital health innovation, driving trust, investment, and adoption at scale.

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

Today, the most readily available and valuable health data to patients and caregivers includes:

- Medication history (from EHR portals and pharmacy apps)
- Basic lab results (via lab portals like LabCorp or Quest)
- Vaccination records
- Appointment summaries and diagnoses (from patient portals like MyChart)
- CMS claims summaries via tools like Blue Button 2.0

While useful, this data is often scattered, out of sync, and not delivered in a format that supports proactive care navigation or longitudinal health understanding, particularly for high-risk or underserved populations.

a. What data is valuable but hard to access?

Many of the most clinically and contextually important data types remain inaccessible to patients, caregivers, and developers. These include:

- Encounter notes and discharge summaries, especially from inpatient stays or out-of-network providers
- Radiology images and reports
- Pathology and genomic test results
- Social determinants of health (SDOH) data collected by community or Medicaid programs
- Real-time appointment availability across multiple provider systems
- Prior authorization status and real-time benefits (RTB) data
- Operative reports and detailed procedural documentation
- Pricing information for upcoming services or medications

The primary barriers are:

- Lack of mandatory TEFCA participation by data holders (especially specialty labs, imaging centers, and commercial payers)
- Limited FHIR API adoption among EHR and payer systems

- No shared patient identity credential to unify data across sources
- Reluctance from vendors to share unstructured or semi-structured data
- Consent complexity for caregivers or patients managing multiple dependents

b. What other sources, beyond claims and clinical data, are of highest value?

Key non-clinical data sources include:

- SDOH data (e.g., food insecurity, housing instability, transportation access) – critical for personalizing care plans and prioritizing outreach in underserved communities.
- Remote monitoring and wearable device data – provides real-time insights into patient activity, vitals, and trends between visits.
- Community-based services data – referrals and outcomes from CBOs, social workers, or health navigators.
- Pharmacy benefit manager (PBM) data – including formulary alignment, prior authorization, and out-of-pocket costs.
- Patient-reported outcomes and goals of care, often overlooked in Electronic Health Records (EHRs), are crucial for delivering value-based and personalized care.

When integrated, these sources provide a 360-degree view of the patient, supporting more equitable and targeted care strategies.

c. What opportunities and challenges exist to improve integration across sources?

Opportunities:

- Mandating TEFCA participation for all CMS-contracted entities, including labs, imaging facilities, and payers, would unlock bidirectional access to both structured and unstructured data.
- Standardizing CSP-based digital identity would enable data aggregation across platforms with verified consent, thereby enhancing user privacy and security.
- Expanding USCDI to include SDOH, device, and patient-reported data elements would improve semantic interoperability.
- Encouraging participation in shared data registries or TEFCA analytics nodes could support research and real-time insights across large populations.

Challenges:

- Data fragmentation across proprietary systems and inconsistent adherence to FHIR standards.
- Lack of incentives for providers or vendors to expose full encounter content or imaging.
- Privacy concerns around sharing SDOH or behavioral health data without proper controls
- Patient matching errors often result from missing or mismatched identifiers in legacy systems.
- Lack of transparent governance on who can act on, aggregate, or enrich multisource data under federal programs.

To enable meaningful clinical research, innovation, and equitable care, CMS should strengthen mandates around TEFCA participation, FHIR API exposure, and digital identity, creating a seamless, secure, and patient-authorized data layer across the healthcare ecosystem.

PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?

To maximize its value to patients, caregivers, and developers, the Blue Button 2.0 API should evolve from a claims-only API into a longitudinal health data platform. This requires incorporating real-time and clinical data elements, not just retrospective billing data. Developers consistently cite the following additional data types as most valuable:

Additional CMS data sources needed:

- Real-time eligibility and plan benefits data, including coverage status, cost-sharing amounts, and out-of-pocket estimates
- Prior authorization status and requirements
- Social Determinants of Health (SDOH) indicators available through Medicaid or Medicare Advantage assessments (e.g., transportation needs, housing status)
- Prescription drug formulary and utilization data, including adherence flags
- Immunization records and gaps in care
- Risk adjustment factor scores (RAF/HCC) and attributed provider information
- Care plans and transition-of-care documentation
- Provider quality scores and network participation

a. What difficulties are there in accessing or utilizing these data sources today?

- Many of these data sources are scattered across CMS systems and not exposed via APIs.
- Some datasets (e.g., prior authorization or SDOH data) are held by Medicare Advantage or Medicaid Managed Care Organizations (MCOs), which are not mandated to expose them in standardized formats.
- There is no standardized authentication or consent process in place to allow third-party applications to retrieve this data on behalf of patients, particularly for caregivers managing dependents.
- Data latency is a significant issue; claims data lags by weeks or months, limiting its utility for real-time decision-making.

b. What suggestions do you have to improve the Blue Button 2.0 API experience?

1. Migrate to a fully FHIR-native architecture, incorporating the latest USCDI versions and supporting Subscriptions and Bulk FHIR for more dynamic queries.
2. Enable real-time patient identity and consent via CSPs (e.g., ID.me, Login.gov).
3. Include clinical and SDOH data from Medicare Advantage encounters and Medicaid assessments.

4. Standardize plan benefit data access through a unified "Digital Formulary and Coverage API".
5. Allow third-party apps to query longitudinal data across CMS programs, including hospice, home health, and ACO attribution, to facilitate seamless integration.

c. Is there non-CMS data that should be included in the API?

Yes. CMS should partner with TEFCA QHINs to ingest and expose the following through Blue Button APIs:

- Lab results and imaging reports from commercial labs (e.g., LabCorp, Quest, BioReference).
- Pharmacy fill data from PBMs.
- Home health and DME utilization data from ancillary providers.
- Community-based services and referrals from local aging and social support agencies.
- Behavioral health encounter data, when consented.

The Blue Button 2.0 API should evolve into a TEFCA-aligned, longitudinal health API, enabling patient- and caregiver-facing tools to deliver personalized, real-time guidance across clinical, financial, and social domains. CMS is uniquely positioned to orchestrate this transformation by mandating participation, unifying APIs, and integrating modern identity and consent standards.

PC-10. How is TEFCA currently helping to advance patient access to health information in the real world?

a. Specific Examples of Impact

TEFCA is beginning to enable real-world patient access by operationalizing national data exchange through designated Qualified Health Information Networks (QHINs). For example:

- At Health Gorilla, one of the first designated QHINs, the Patient360 product has enabled patient-authorized access to their longitudinal clinical histories across multiple health systems, including electronic health records (EHRs), labs, and pharmacies.
- beHuman Corp., built on this infrastructure, leverages TEFCA to pull clinical records for cancer screening eligibility, even for patients without a primary care provider, improving access for underserved populations.
- Early integrations with providers and developers have leveraged TEFCA to reduce redundant testing, surface prior conditions, and close care gaps more efficiently.

These examples demonstrate that TEFCA offers scalable, consent-driven data access, enabling it to reach patients who have been previously excluded from the digital health ecosystem.

b. Recommended Changes to Improve TEFCA's Impact

1. Mandate TEFCA participation for all CMS-contracted payers, major labs (e.g., LabCorp, Quest), and certified EHR vendors. Voluntary participation limits completeness.

2. Enable patient-directed queries through certified third-party apps, not just provider-driven queries.
3. Standardize digital identity workflows using NIST 800-63-3 IAL2 CSPs (e.g., Login.gov) to support seamless consent.
4. Embed TEFCA APIs in CMS Blue Button infrastructure, allowing patients to access both claims and clinical data in one interface.
5. Fund developer-facing sandboxes and bulk-data testbeds to accelerate innovation.

c. High-Impact Use Cases for TEFCA Implementation

- Preventative care eligibility checks (e.g., cancer screening, vaccines) based on complete patient history
- Cross-provider medication reconciliation at the point of care
- Real-time ADT notifications across state lines and unaffiliated systems
- ACO and value-based care attribution using longitudinal data
- Proxy access for caregivers managing multiple patients with verified credentials
- Patient self-navigation tools to show gaps in care and covered services
- Digital prior authorization pre-checks, powered by payer + provider data

d. Effective Standards in Current Use

- FHIR R4 for data structuring and patient-facing APIs
- OAuth 2.0 and OpenID Connect for secure authentication
- C-CDA for backward-compatible document exchange
- IHE XCA/XDS.b for cross-community queries in HIE settings
- SMART on FHIR for app authorization and launch within EHRs

e. Underutilized Standards That Could Improve Access

- FHIR Subscriptions for real-time updates and event-driven workflows
- CDS Hooks to surface context-aware recommendations during care
- Bulk FHIR (Flat FHIR) for population-level insights and multi-patient navigation
- UDAP (Unified Data Access Profiles) for secure, scalable API authorization
- FHIR Questionnaire and QuestionnaireResponse for dynamic intake and assessments

f. Redundant Channels That Should Be Consolidated

- DirectTrust/Direct Secure Messaging, which is often siloed and poorly integrated with patient apps
- Proprietary EHR APIs, which vary widely and undermine developer scalability
- Point-to-point HL7v2 interfaces, which are costly to maintain and lack patient-facing transparency

All of these should be migrated to TEFCA-based exchange using modern APIs and common security standards.

g. Are There Adequate Alternatives to TEFCA?

No existing alternative matches TEFCA's national governance, trust framework, and scalability. Some alternatives are:

- DirectTrust lacks standardization and consumer scalability.
- Private EHR networks, such as Epic Care Everywhere, are fragmented and proprietary.
- Regional HIEs offer value but are limited by geography.

Only TEFCA is positioned to deliver a nationwide, consent-driven, multi-party data exchange that includes patients, payers, providers, and public health entities.

TEFCA is the backbone of equitable, national-scale data exchange. With stronger policy levers—mandates for key data holders, integration into CMS programs, and developer support—it can unlock the next generation of patient-centered innovation.

PC-11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?

a. Value, Availability, and Accuracy of HIE Data

HIEs provide critical infrastructure for enabling patient access to health data, particularly in regions without robust EHR vendor networks or national QHIN coverage. The value of HIE data is highest when:

- It includes comprehensive longitudinal records (e.g., labs, ADTs, medications, imaging reports)
- It is timely, with real-time feeds from hospitals and large outpatient systems
- It supports event-based notifications (e.g., hospital admissions and discharges)

However, data availability and accuracy vary widely across HIEs. Some operate only at the encounter level without full context or structured metadata. Others lack payer data, SDOH inputs, or behavioral health records. Patient identity matching is often inconsistent, leading to duplication or fragmentation.

b. Recommended Changes to Improve HIE Performance

1. Standardize data formats across HIEs using USCDI and FHIR to ensure consistency and semantic interoperability.
2. Mandate bidirectional patient access via open APIs for any HIE that receives federal funds or is recognized under TEFCA.
3. Align HIEs with TEFCA participation, so data shared within regional networks becomes available nationally via QHINs.
4. Require HIEs to adopt CSP-based identity verification to support secure patient access and cross-HIE data portability.

5. Expand funding for HIE integration with community-based organizations (e.g., FQHCs, social services), especially in Medicaid-heavy regions.

c. High-Performing HIE Models to Replicate

Several HIEs stand out for their governance, data services, and patient access models:

- CRISP (Maryland): Provides ADT alerts, prescription monitoring, and quality reporting integration for both providers and patients.
- Indiana Health Information Exchange (IHIE): Offers a highly integrated longitudinal patient record used by multiple health systems across Indiana.
- Manifest MedEx (California): Aggregates data from providers, payers, and public health agencies and exposes it through FHIR-based APIs.

These HIEs succeed by maintaining broad participation agreements, governance transparency, and multi-stakeholder collaboration, including patient-centered services.

d. Ongoing Role of HIEs in a TEFCA Era

HIEs will remain vital, but their role must evolve. Rather than operating as isolated networks, HIEs should serve as regional data hubs that plug into the TEFCA framework. Their comparative advantages include:

- Local trust relationships.
- Expertise in data normalization and quality control.
- Support for regional use cases (e.g., Medicaid, disaster response, behavioral health integration).

As TEFCA expands nationally, HIEs should:

- Become a TEFCA participant or subparticipant.
- Act as intermediaries between QHINs and local systems.
- Provide value-added services, such as analytics, quality reporting, or SDOH enrichment, in addition to the standardized TEFCA exchange.

HIEs should complement, not compete with, TEFCA, offering local depth while supporting national reach.

HIEs remain essential building blocks of interoperability, but to serve patients fully, they must be aligned with national standards, governed transparently, and integrated into TEFCA. CMS should encourage this alignment through funding incentives, certification pathways, and TEFCA onboarding mandates tied to participation in federal programs.

PC-12. What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both?

a. Key High-Value Use Cases:

The most valuable operational use cases include:

1. Binding Cost Estimates for Pre-defined Periods
    - Patients should be able to request procedure-specific, benefit-adjusted cost estimates that remain valid for 30 to 90 days.
    - Use case: A Medicare beneficiary evaluating colorectal cancer screening options should see total out-of-pocket costs for in-network providers, factoring in deductibles and copays.
2. Viewing Real-Time Provider Schedule Availability
    - Patients should be able to view appointment slots across providers via APIs, facilitating same-day or next-available scheduling.
    - This would reduce call volume, lower no-show rates, and create more competitive access to care.
3. Third-Party App Integration for Appointment Management
    - Tools like Zocdoc are limited to specific providers. CMS could help standardize appointment APIs (e.g., FHIR Scheduling) to democratize this feature.
    - This would empower developers to build unified care navigation apps, thereby reducing friction for patients who manage multiple providers.
4. Access to Provider-Level, Patient-Facing Quality Metrics
    - Metrics such as complication rates, readmission rates, and patient satisfaction should be standardized and queryable via an API.
    - This increases transparency and encourages provider competition based on outcomes, not geography.
5. Intelligent Provider Search Based on Personalized Needs
    - A recommendation engine combining language preference, insurance network, specialty, proximity, and SDOH accommodations.
    - Example: A caregiver for a Spanish-speaking senior could find providers accepting Medicare and offering in-home follow-ups in zip code 33160.

b. Use Cases Possible Today:

- Partial cost estimate tools through insurer portals (often outdated or generic).
- Third-party appointment booking apps, however, have incomplete coverage.
- CMS Care Compare offers quality ratings, but its data is limited to the facility level.
- Health plan directories are often poorly maintained and not accessible via application programming interfaces (APIs).

These tools are fragmented and not easily composable into a single, seamless user experience.

c. What Should Be Possible in the Near Future:

- FHIR APIs for real-time provider schedules are being adopted across certified Electronic Health Record (EHR) systems.

- Standardized provider directory APIs including digital identity verification and FHIR endpoint discovery.
- CMS mandates the use of API-based cost estimation services for all Medicare Advantage and Qualified Health Plan (QHP) plans.
- Integration of CMS and TEFCA data in third-party apps, allowing navigation tools to combine claims, clinical history, and benefit information.

This would support whole-person, real-time navigation by any patient-facing tool.

d. What Would Be Very Valuable, but May Be Hard to Achieve:

- Real-time bundled episode pricing with provider-specific commitments and quality guarantees.
- Automated, personalized care path recommendations, integrating patient preferences, clinical risk scores, social needs, and real-time availability.
- Cross-payer, cross-provider care scheduling orchestration, with AI agents comparing options and guiding patients through next steps—across systems and regions.

These use cases require deep interoperability, harmonized governance, and regulatory mandates for participation, particularly from commercial payers and specialty providers.

To unlock these operational use cases, CMS should mandate the adoption of FHIR Scheduling APIs, incentivize the use of price transparency APIs, and align payer and provider participation through TEFCA and CSP-based identity protocols. These moves would empower patients with tools that simplify navigation, increase transparency, and foster meaningful competition in healthcare delivery.

PC-13. How can CMS encourage patients and caregivers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would increasing the reporting of complaints advance or negatively impact data exchange?

CMS can take several key actions to significantly improve awareness, usability, and impact of the information blocking complaint process among patients and caregivers:

1. Embed Reporting Tools Directly into Patient Portals and Medicare Systems

- Add a "Report a Data Access Issue" button within Medicare.gov, Blue Button-enabled apps, and patient portals of TEFCA-participating organizations.
- Use SMART on FHIR launch contexts to pre-fill the complaint form with relevant metadata (e.g., organization, patient ID, date of attempt, type of denial).

2. Fund Outreach and Awareness Campaigns

- CMS should run targeted educational campaigns through 1-800-MEDICARE, community health workers, and Medicare Advantage plan mailings to educate patients about:

    - Their right to access data
    - What constitutes information blocking
    - How and where to file a complaint

This is especially important for underserved populations, who are most likely to experience silent forms of blocking.

3. Incentivize Reporting via Trusted Third Parties

- Allow QHINs, digital health app developers, or consumer advocacy groups to assist in submitting reports with patient consent.
- Provide legal protections for whistleblowers who identify systemic blocking practices.

4. Improve the Portal Interface and Feedback Loop

- Redesign the Information Blocking Portal for accessibility and multilingual use.
- Create a complaint tracking dashboard that allows patients to track the progress and outcomes of their submissions.

Impact of Increased Reporting:

- Positive Effects:
    - Drives accountability among providers, EHR vendors, and payers.
    - Highlights systemic patterns of non-compliance, which can inform policy and enforcement priorities.
    - Empowers patients and builds public trust in CMS and ONC efforts.
- Risks to Mitigate:
    - Risk of frivolous or misinformed submissions can be mitigated through clear in-portal guidance and filters.
    - Some stakeholders may fear reputational harm, but this concern can be mitigated through the use of anonymized public dashboards and established due process protocols.

Encouraging and simplifying the submission of information blocking complaints is a necessary enforcement lever that strengthens TEFCA's utility, supports patient rights, and incentivizes data holders to comply with interoperability rules. Far from hindering data exchange, increased reporting will advance the culture of data transparency and equity that TEFCA and CMS are working to build.

PC-14. Digital Identity Credentials (e.g., CLEAR, Login.gov, ID.me, etc.)

a. Challenges in Getting Patients/Caregivers to Use Digital Identity Credentials:

- Low awareness: Many patients and caregivers are unaware that digital identity credentials even exist or why they matter.

- Complex onboarding: Verifying identity through CSPs can involve uploading documents, taking selfies, or using mobile apps—steps that are difficult for elderly, disabled, or non-English-speaking users.
- Limited digital literacy: A significant portion of the Medicare population lacks experience with two-factor authentication, smartphone apps, or account recovery methods.
- Trust concerns: Patients may distrust government or commercial identity providers due to concerns about privacy or past data breaches.

b. Benefits of Widespread Use:

- Seamless access to personal health records across payers, providers, and apps with one secure login.
- Proxy access for caregivers, with validated relationships and audit trails.
- Secure consent management for third-party data sharing.
- Reduced fragmentation: Eliminates the need for multiple usernames/passwords across different patient portals.
- Supports TEFCA exchange: Enables real-time patient-authorized access to longitudinal records.

c. Potential Downsides:

- Digital divide: Without inclusive onboarding options, some patients may be excluded.
- Privacy anxiety: Concerns about surveillance, data tracking, or identity misuse could slow adoption.
- Credential loss: Forgotten passwords or inaccessible devices can prevent patients from accessing their medical information, especially in emergency situations.

d. How CSP Use Improves Access:

- Enables single-sign-on (SSO) to all TEFCA-connected apps and portals using strong, standardized authentication.
- Facilitates auditability and accountability for patient-authorized data exchanges.
- Streamlines real-time consent workflows, reducing friction on the provider side.
- Simplifies caregiver delegation, proxy management, and guardianship workflows across health systems.

e. Role of Stakeholders in Driving Adoption:

- CMS should:
    - Mandate CSP acceptance for certified apps and Medicare-facing systems.
    - Provide grants for FQHCs and rural clinics to support in-person identity verification.
    - Establish digital identity as a core part of Meaningful Use and TEFCA onboarding requirements.
- Payers should:

    - Auto-enroll new members in CSP workflows during benefits onboarding.
    - Embed CSP credentials into member portals and mobile apps.
- Providers should:
    - Enable CSP login for patient portals.
    - Offer in-clinic enrollment support.
    - Educate patients about the benefits of digital identity at the point of care.
- App Developers should:
    - Integrate federated CSP login into onboarding.
    - Build credential recovery and multilingual support into user interfaces.

f. How CMS Can Encourage Adoption:

- Bundled enrollment: Offer digital identity setup as part of Medicare registration (online or via 1-800-MEDICARE).
- Multilingual education campaigns: Run ads, mailers, and videos explaining how digital identity enables easier access and better care.
- Credential subsidies: Offer financial or technical assistance for CSP registration in underserved areas.
- Incentives: Link digital identity usage to faster prior authorization, simplified eligibility checks, or bonus payments under value-based models.

Digital identity is the gateway to patient-centered interoperability. By addressing onboarding friction, building trust, and embedding CSPs into all CMS and provider systems, we can unlock seamless, secure, and scalable access to health data for all beneficiaries and caregivers.

PROVIDERS (PR)

1. DIGITAL HEALTH APPS

PR-1. Encouraging Providers to Leverage Approved Digital Health Products

CMS can significantly accelerate provider adoption, especially in rural and underserved areas, by aligning financial incentives, reducing technical barriers, and simplifying compliance pathways for integrating CMS-approved digital health products.

Key recommendations include:

- Tie adoption to reimbursement incentives under MIPS, ACO REACH, and MSSP.
- Subsidize onboarding and offer technical assistance for small practices and rural health centers.
- Create a CMS-backed certification directory of digital tools with demonstrated usability, equity impact, and TEFCA integration.
- Embed digital navigation tools into certified EHR workflows to reduce administrative overhead and drive patient engagement.

a. Current Obstacles:

- Limited broadband and infrastructure in rural areas impairs reliable access to digital tools.
- EHR vendor lock-in and lack of integration capabilities make adoption costly and complex.
- Workforce shortages reduce the capacity to train staff or maintain new tech.
- Lack of reimbursement for time spent using or managing digital tools (e.g., remote engagement or care coordination).
- Concerns about liability and privacy compliance when using third-party tools.

b. Information Providers Should Share with Patients:

When introducing digital health products, providers should clearly explain:

- What the tool does (e.g., manage medications, schedule appointments, access records).
- What data is accessed or shared, and who can view it.
- How privacy is protected, including the use of TEFCA and NIST-compliant identity protocols.
- How the tool supports the care plan or improves health outcomes.
- Whether use is optional, and how to opt in or out.

This transparency fosters trust and informed decision-making.

c. Provider Responsibilities When Recommending Digital Tools:

Providers who recommend digital products must:

- Ensure tools are CMS-approved or ONC-certified, with validated clinical and privacy protections.
- Assess patient readiness, including digital literacy and language access.
- Document clinical use and relevance in the medical record (e.g., if used for risk assessment or care planning).
- Support patient onboarding, especially for elderly or underserved individuals.
- Monitor outcomes and report adverse events or usability barriers when applicable.

By embracing these responsibilities, providers help ensure digital health tools contribute to safe, equitable, and effective care delivery.

CMS should foster a supportive ecosystem that enables rural and underserved providers to receive financial rewards, technical support, and legal protection when adopting digital health tools. This includes embedding TEFCA as the data backbone and digital identity as the front door, ensuring no patient is left behind in the transition to connected care.

PR-2. Obstacles to the Development and Utilization of Digital Tools for Physician Workflows

Key Obstacles:

1. Lack of Standardized Data Models and APIs
    - Quality measurement and billing rely on fragmented, non-standardized data, such as scanned PDFs and unstructured notes.
    - Inconsistent FHIR implementations across Electronic Health Records (EHRs) make the development of third-party apps challenging.
    - Vendors often require proprietary APIs or custom integrations, increasing complexity and cost.
2. Vendor Lock-In and Interoperability Barriers
    - Many EHR vendors resist seamless third-party integration to preserve their competitive advantage.
    - High interface fees, closed development environments, and a lack of transparent documentation deter innovation.
    - CDS Hooks, SMART on FHIR, and Bulk FHIR are underused due to weak enforcement.
3. Redundant and Manual Quality Reporting Processes
    - Providers must submit data to multiple payers and CMS programs using different formats and timelines.
    - QRDA remains complex and duplicative when compared to emerging Bulk FHIR-based reporting.
    - Manual abstraction still dominates, especially for smaller practices.
4. Insufficient Real-Time Data Access
    - Clinical documentation tools often lack real-time access to TEFCA-exchanged data, such as laboratory results, consultation notes, and advance directives.
    - Billing systems often can't validate eligibility or authorizations during patient intake.
5. Regulatory Uncertainty and Liability Concerns
    - Fear of violating HIPAA or encountering information blocking allegations discourages the use of innovative tools.
    - Providers are unsure of their responsibilities when using AI-assisted documentation or digital quality calculators.

Mitigation Strategies:

1. Mandate Open, Certified APIs for Workflow Integration
    - Require EHRs to support CDS Hooks, FHIR Scheduling, Prior Auth APIs, and TEFCA queries within core workflows.
    - Incentivize vendors to publish developer-friendly APIs and sandbox environments.
2. Replace QRDA with FHIR-Based Quality Reporting
    - CMS should transition to Bulk FHIR exports for eCQM data collection and allow centralized registry submission.

    - Enable CMS to calculate measures from raw data, thereby reducing provider workload.
3. Enable TEFCA-Based Access Within Workflow Tools
    - Require EHRs and digital health apps to use TEFCA for clinical document retrieval, reconciliation, and summarization.
    - Embed TEFCA-powered clinical summaries directly into documentation templates and billing engines.
4. Fund Open-Source Workflow Tools for Small Practices
    - CMS should support the creation of open-source quality measure calculators, billing reconciliation tools, and intake modules built on FHIR.
    - Provide grants to integrate these into certified Electronic Health Records (EHRs) used by rural and safety-net providers.
5. Provide Legal and Compliance Guidance
    - CMS and ONC should issue formal guidance on the safe and appropriate use of third-party AI, clinical decision support (CDS), and quality tools.
    - Clarify when app usage constitutes patient-directed access vs. provider-initiated workflows.

To unleash innovation in physician workflows, CMS must align incentives, mandate standards-based APIs, and reduce reliance on legacy formats. TEFCA should serve as the data highway, and CMS-certified FHIR APIs should become the on-ramps, ensuring that all providers, regardless of size, can benefit from intelligent, interoperable, and automated digital tools.

PR-3. Importance of Accessing All EHR Data Formats for Exchange

Ensuring that *all* data within an Electronic Health Record (EHR), regardless of format, is accessible and exchangeable is essential for delivering equitable, high-quality healthcare in both urban and rural settings. Without access to scanned documents, faxed records, narrative notes, and structured fields alike, providers lack the complete clinical picture needed to make timely, safe, and cost-effective decisions.

a. Current Challenges in Accessing Different Data Formats:

- Scanned documents (e.g., paper referrals or outside test results) are often stored as images or PDFs with no searchable or computable content.
- Faxed records are inconsistently indexed, frequently misplaced, and cannot be parsed without manual intervention.
- Free-text notes contain valuable clinical insights (e.g., symptom progression, rationale for treatment), but they are not consistently extracted or shared.
- Even structured fields, such as medication lists or problem summaries, may not follow consistent vocabularies (e.g., RxNorm, SNOMED), which can hinder interpretation by external systems.

b. Impact on Patient Care Quality:

- Missed diagnoses or duplicative testing when prior imaging or lab results are buried in inaccessible formats.
- Medication errors occur when allergy lists or reconciliations are locked in unstructured discharge summaries.
- Delayed treatment due to inability to verify prior authorizations, external consult notes, or recent procedures.
- Poor coordination, especially during transitions of care, occurs when providers lack a full view of the patient's clinical history.

These issues disproportionately affect rural areas where patients frequently cross institutional boundaries and providers rely on partial or delayed records.

c. Technical Barriers to Full Data Accessibility:

- Legacy EHR systems lack OCR, NLP, or API infrastructure to convert and expose non-structured data.
- Vendor reluctance to open up document imaging archives or enable indexing of free-text notes.
- Lack of standardization for labeling and categorizing scanned or non-coded documents (e.g., which documents to include in a CCD or FHIR `DocumentReference`).

d. Cost or Privacy Implications:

- Costs include retrofitting legacy systems, deploying NLP/OCR tools, and performing QA to ensure accurate extraction.
- Privacy implications arise from broader data availability, including an increased risk of PHI exposure from narrative notes or scanned referrals that contain sensitive details.
- These concerns are addressable through:
    - Fine-grained access control policies,
    - Audit logging, and
    - Use of CSP-based digital identity to enforce consent and authorization boundaries.

e. Priority Level Compared to Other Interoperability Needs:

This issue is high priority, particularly for safety-net, rural, and community-based providers who often rely on incomplete records. While TEFCA and FHIR APIs are enabling the exchange of structured data, full interoperability will remain elusive until *all clinically relevant data*, regardless of format, is accessible and usable.

CMS should:

- Mandate inclusion of scanned and free-text content in TEFCA exchange payloads (e.g., via DocumentReference and DiagnosticReport resources).
- Incentivize EHR vendors to implement OCR and NLP pipelines to surface insights from unstructured data.

- Require certified health IT systems to make *entire patient records* exportable and queryable, including non-coded attachments.

Access to all EHR data formats is not a luxury—it is a clinical necessity. Unlocking this data through TEFCA, combined with identity-driven consent models and modern data transformation tools, will elevate interoperability from a regulatory requirement to a true enabler of high-quality, equitable care nationwide.

PR-4. Enabling Access to Administrative Workflows for Third-Party Digital Products

Key Barriers Today:

Most administrative functions—such as completing intake forms, scheduling appointments, and accessing the provider directory—are still locked behind proprietary, siloed interfaces controlled by Electronic Health Record (EHR) vendors or practice management systems. This prevents third-party apps from delivering seamless, consumer-grade healthcare experiences and limits the automation and patient engagement capabilities.

Recommended Changes and Improvements:

1. Expand API Certification Requirements

- Mandate standardized FHIR APIs under the ONC Health IT Certification Program for the following workflows:
  - Appointment scheduling (FHIR `Appointment` + `Slot`)
  - Patient intake and registration (using `Questionnaire`, `QuestionnaireResponse`)
  - Insurance eligibility checks (via `Coverage` and `EligibilityRequest/Response`)
  - Provider directory search (including FHIR `PractitionerRole`, `Location`, and digital ID integration)
- These APIs must be made available to third-party applications, not just internal modules.

2. TEFCA Must Support Administrative Data Exchange

- Administrative workflows should be explicitly included in TEFCA's scope, enabling the trusted exchange of scheduling, demographics, payer info, and digital identities.
- CMS could require all TEFCA QHINs to support administrative resources (not just clinical documents) and to make these available via query.

3. Require OAuth2 + SMART on FHIR Authorization

- Patients and caregivers should be able to authorize access to their scheduling and intake data using OAuth2 tokens through their chosen digital health apps.
- Apps should be able to read/write appointment and intake data once authorized, eliminating paper-based, error-prone processes.

4. Incentivize Provider Participation

- CMS should tie MIPS/APM incentives to provider participation in certified scheduling and intake application programming interfaces (APIs).
- Offer additional incentives for rural or resource-constrained clinics that onboard open-access appointment APIs.

5. Promote Uniform Data Models and Directories

- CMS should maintain or support a national FHIR-based provider directory, as described in PA-4, with linked digital identity credentials.
- Each provider's available services, language capabilities, and schedule slots should be structured and queryable through open APIs.

6. Clarify Regulatory Support and Liability Protections

- CMS and ONC should publish guidance clarifying that third-party app access to administrative workflows is covered under HIPAA and information blocking rules and is not restricted to provider portals.
- Offer liability protections for providers that integrate open-access administrative APIs in good faith.

To bring healthcare in line with other consumer-facing industries, administrative interoperability must be treated as a first-class priority alongside clinical data sharing. CMS should expand certification, enforce open standards, and mandate TEFCA alignment to enable a new generation of digital products that simplify patient intake, scheduling, and engagement, particularly for underserved populations.

PR-5. Supported FHIR APIs and Capabilities at beHuman Corp.

a. Patient Access API

- Transaction model: Individual queries using OAuth2-authenticated FHIR calls.
- Use case: Enables patients to retrieve their clinical summaries, lab results, medication history, and claims data through the beHuman app, leveraging TEFCA-connected QHINs for data aggregation.
- Constraints: Some upstream data holders do not yet support full USCDI payloads or enforce patient identity revalidation, which can result in delayed access.

b. Standardized API for Patient and Population Services

- Transaction model: Both individual and bulk queries (via Group-based operations).
- Use case: Used to run population-level screening campaigns (e.g., opted-in patients eligible for cancer screening based on age, risk, and history) and to push clinical reminders or care gap alerts. Currently, beHuman relies on HRSA-identified geographies for targeted engagement with underserved populations.

- Constraints: Dependent on payer or EHR partner capabilities to support bulk endpoints and standardized group attribution logic.

c. Provider Directory API

- Transaction model: Read access via FHIR PractitionerRole, Location, and Organization resources.
- Use case: Used to verify provider credentials, filter referral networks, and support patient-provider matching during scheduling.
- Constraints: Many provider directories are incomplete or not updated in real time; CMS-backed FHIR directory adoption would improve consistency.

d. Provider Access API

- Transaction model: Patient-mediated, with provider authorization enabled through Qualified Health Information Networks (QHINs).
- Use case: Used by licensed beHuman providers to access external clinical records (e.g., past colonoscopies, mammograms, or chronic condition history) before telehealth screenings.
- Constraints: TEFCA-based access only works if counterpart systems are QHIN-connected; legacy providers still rely on fax or Direct messaging.

e. Payer-to-Payer API

- Transaction model: Planned support using FHIR-based exchange for prior coverage history.
- Use case: This will enable continuity of care assessments and eligibility checks during the initial intake, particularly for new Medicare beneficiaries switching to a new plan.
- Constraints: Payer APIs are inconsistently implemented; CMS should mandate standardization under TEFCA.

f. Prior Authorization API

- Transaction model: Currently in development using FHIR `PriorAuthorization` and `CoverageRequirementsDiscovery`.
- Use case: Will automate preauthorization for cancer screening labs, reducing friction and improving test turnaround time.
- Constraints: Widespread implementation across labs and payers remains lacking; regulatory mandates are necessary for consistent support.

g. Bulk FHIR

- Transaction model: Group ID–based bulk data exports.
- Use case: Used for identifying eligible screening populations, generating longitudinal health snapshots, and exporting structured data for CMS quality reporting.

- Constraints: Some upstream systems lack scalable Bulk FHIR endpoints or struggle with identity resolution across patient panels.

h. SMART on FHIR

- Transaction model: Both EHR-launched and standalone apps are supported.
- Use case: Enables single sign-on and launch of beHuman screening tools directly from within Electronic Health Records (EHRs) or mobile devices. OAuth 2.0 is used to authorize and retrieve the patient context.
- Deployment process: This involves registering the app with the EHR vendor or health system, obtaining client credentials, and configuring launch context mappings.
- Constraints: App approval varies by EHR vendor; lack of centralized registration slows scaling.

i. CDS Hooks

- Transaction model: Context-triggered real-time decision support.
- Use case: Integrated with intake, telehealth, and test ordering workflows to suggest screenings based on risk factors or gaps in care (e.g., missed mammogram for an eligible patient).
- Constraints: Limited support across vendor platforms; often requires custom implementation unless supported natively by the host EHR.

beHuman's architecture is designed to leverage the full potential of modern FHIR-based APIs, TEFCA connectivity, and AI-driven decision support to streamline cancer screening and preventative care. Broader, mandated adoption of these standards—especially Prior Auth, Bulk FHIR, and CDS Hooks—would unlock even greater clinical and operational efficiencies.

PR-6. TEFCA and Provider Access to Health Information

a. Is TEFCA currently helping to advance provider access to health information?

Yes. TEFCA is starting to demonstrate real value for provider data access by establishing a standardized, trusted framework for the cross-network exchange of longitudinal clinical records. Providers connected through QHINs (Qualified Health Information Networks) can now query external sources in real-time, eliminating the need for faxes, Direct messages, or multiple patient portals.

Examples include:

- BeHuman providers, through integration with Health Gorilla (a Qualified Health Information Network, or QHIN), can retrieve complete clinical histories during intake, including lab results, imaging summaries, and prior diagnoses from unaffiliated health systems, enabling same-day screening decisions.

- The TEFCA exchange has supported medication reconciliation workflows across providers, pharmacies, and Electronic Health Records (EHRs) in Medicare Advantage populations, improving prescribing safety during transitions of care.

b. What changes would you suggest?

To realize TEFCA's full provider-enabling potential, CMS and ONC should:

- Mandate TEFCA participation for all CMS-contracted entities (e.g., labs, imaging centers, MA plans, Medicaid MCOs).
- Require EHR vendors to embed TEFCA query capabilities natively into clinical workflows (e.g., as a tab in the chart, not hidden in APIs).
- Fund onboarding for small and rural practices to integrate with TEFCA through certified intermediaries.
- Standardize response times and payload formats across QHINs to reduce variability and improve reliability.

c. What other options are available outside of TEFCA?

While TEFCA is becoming the national backbone, other mechanisms still exist:

- **Carequality and CommonWell** offer peer-to-peer interoperability for select Electronic Health Record (EHR) vendors; however, they lack federal governance and uniform participation rules.
- **HIEs** offer regional or state-level data exchange, but with inconsistent coverage and varying technical standards.
- **Point-to-point APIs** and **Direct Messaging (via DirectTrust)** remain in use but are fragmented, duplicative, and lack scale.

None of these options offers the compliance, universality, or trust framework that TEFCA is positioned to deliver.

d. Are there redundant standards, protocols, or channels that should be consolidated?

Yes. The following should be consolidated or phased into TEFCA:

- Direct Messaging (SMTP/XDR): Used inconsistently, often duplicative, and challenging to track across care settings.
- Custom vendor APIs: Proprietary interfaces for record retrieval or appointment exchange should be replaced with FHIR R4 or C-CDA over TEFCA-compliant endpoints.
- Overlapping regional HIE protocols: Should transition to become subparticipants under TEFCA QHINs, rather than maintaining standalone governance or transport mechanisms.

By consolidating under TEFCA, CMS can reduce the administrative burden on providers, improve data consistency, and support a scalable, national interoperability model.

TEFCA is advancing provider access to health information; however, its momentum must be accelerated through mandatory participation, workflow integration, and the elimination of redundant exchange models. With these policy changes, TEFCA can become the universal operating layer for real-time, secure, and equitable access to health data.

PR-7. Strategies to Support Provider Interoperability and Reduce Burden

1. Align Interoperability with Existing Incentives and Metrics

CMS should integrate interoperability requirements directly into value-based care frameworks, MIPS scoring, and APM participation, rather than creating separate compliance tracks. For example:

- Reward providers for submitting clinical quality measures via Bulk FHIR rather than QRDA.
- Embed TEFCA participation into ACO REACH, MSSP, and MIPS Improvement Activities.
- Use existing chronic care management (CCM) or transitional care management (TCM) workflows to validate and audit real-time data sharing.

This approach turns interoperability from a burden into a performance enabler, reinforcing clinical priorities already tracked by providers.

2. Fund Plug-and-Play Tools and Shared Services

CMS should subsidize the use of shared infrastructure tools that reduce the technical lift for providers, such as:

- Pre-certified FHIR adapters for legacy EHRs.
- Federated credentialing and identity services (e.g., Login.gov integration).
- API management dashboards and data quality monitors are provided through QHINs.

These tools can be developed centrally and deployed to rural or under-resourced providers, requiring minimal to no configuration.

3. Expand Technical Assistance and Regional Support

Establish regional TEFCA onboarding hubs or grants for small and independent practices. These could:

- Offer one-time setup support for TEFCA integration.
- Provide a guide on choosing third-party digital products that are CMS-compliant.
- Serve as liaisons between EHR vendors and QHINs to minimize back-and-forth communication.

This approach helps solve the resourcing gap while promoting uniform national interoperability standards.

4. Reduce Redundancy through Policy Unification

CMS should consolidate overlapping data sharing programs and harmonize standards across:

- Public health reporting
- Quality reporting (eCQMs)
- Risk adjustment submissions
- Patient access APIs

This would allow providers to reuse a single pipeline of FHIR data for multiple reporting and engagement functions.

5. Promote Interoperability-First Certification and Procurement

CMS can drive demand by:

- Requiring ONC-certified, TEFCA-compliant digital products in any CMS-funded program.
- Creating a public registry of certified, low-burden, plug-and-play apps that small practices can adopt with confidence.

This simplifies vendor selection and ensures that innovation aligns with policy goals.

To reduce provider burden while improving interoperability, CMS must reuse existing workflows, fund shared tools, and align incentives. TEFCA should be the default highway, but providers must be equipped with simplified on-ramps and guided by performance-linked benefits, not parallel compliance regimes.

PR-8. Strategies to Simplify Clinical Quality Data Responsibilities

a. Benefits and downsides of using Bulk FHIR data exports from EHRs for CMS quality submissions

Benefits:

- Reduces provider burden by eliminating manual extraction, transformation, and submission of quality data in QRDA format.
- Allows CMS to calculate quality metrics centrally, improving consistency, reducing variation, and avoiding duplicated effort across providers.
- Supports the reuse of structured clinical data for multiple reporting purposes, such as ACO quality, MIPS, and health equity analysis.
- Enables timelier feedback loops to providers on performance trends and care gaps.

Downsides:

- Requires EHR vendors to fully implement and maintain standards-compliant Bulk FHIR endpoints, which may present a technical and financial hurdle.
- Raising potential privacy and security concerns if data scope or access controls are not well-defined.
- Could reduce provider visibility or control over how their performance is calculated unless transparent dashboards and APIs are included.

Recommendation: CMS should provide technical specifications, testing infrastructure (sandbox), and implementation grants to EHR vendors and providers to ensure robust and secure Bulk FHIR implementations.

b. Consolidating interoperability and quality reporting responsibilities

To streamline obligations and increase return on investment, CMS should:

- Unify submission formats: Mandate Bulk FHIR (and SMART on FHIR for individual-level engagement) as the single interoperability and reporting standard.
- Reuse TEFCA-enabled data exchange for both quality measurement and clinical decision support, enabling seamless transition between data capture and reporting.
- Incorporate quality reporting into TEFCA-based query and subscription workflows, reducing the need for batch exports or redundant chart reviews.
- Allow QHINs or certified intermediaries to serve as centralized hubs that both transmit clinical data and return quality dashboards to providers.

This dual-purpose approach simplifies workflow complexity, promotes the structured use of data at the point of care, and aligns with digital-first clinical practices.

c. Requirements for data registries supporting digital quality measurement and clinical care

CMS should require that certified data registries:

- Support FHIR R4 and Bulk FHIR standards, including group-level filtering and query parameterization by provider ID, location, or population cohort.
- Provide real-time, API-based feedback dashboards for providers, surfacing actionable quality performance indicators (e.g., missed screenings, abnormal labs).
- Be connected to TEFCA, ensuring comprehensive data access beyond the reporting provider's EHR.
- Offer auditability, export functionality, and traceable lineage from clinical input to measure output.
- Support stratification by race, ethnicity, SDOH indicators, and insurance status to inform equity-driven interventions.

By embedding these capabilities, registries can evolve from passive repositories to active tools for care improvement and regulatory alignment.

CMS can significantly reduce provider reporting burden and increase quality transparency by:

- Standardizing on Bulk FHIR
- Consolidating interoperability and reporting efforts
- Modernizing data registries to deliver real-time, actionable insights

These changes would support both regulatory compliance and better clinical outcomes in a unified, low-friction digital ecosystem.

3. DIGITAL IDENTITY

PR-9. Encouraging Provider Acceptance of Digital Identity Credentials

a. What would providers need help with to transition to single, trusted digital identity credentials for patients?

Providers will need technical, operational, and financial support to adopt CSP-backed (Credential Service Provider) digital identity systems. Specific areas of assistance include:

- Integration Toolkits: The CMS should fund or provide EHR vendor-neutral "plug-and-play" modules to integrate with CSPs, such as CLEAR, Login.gov, or ID.me. These should work across patient portals, intake systems, and telehealth platforms.
- Workflow Redesign Support: Many providers rely on internal identity systems tied to legacy infrastructure. CMS could fund implementation assistance to help practices streamline login, consent, and proxy workflows using federated credentials.
- Incentives: CMS should tie acceptance of CSP credentials to quality bonus programs, value-based care participation, or TEFCA compliance scoring to drive adoption.
- Patient Education Materials: Providers will also need multilingual, accessible materials to help patients understand and adopt these identity solutions. CMS should offer templated resources that providers can localize.

b. How might CMS balance privacy with convenience and access?

CMS should adopt a privacy-by-design framework that maximizes access without compromising patient trust. Key recommendations include:

- Use Federated Identity Models: Instead of centralized credential storage, CMS should support decentralized, NIST 800-63-3 compliant digital IDs issued by trusted Certificate Service Providers (CSPs). This allows patients to use a single login across all providers without creating a government-run identity repository.
- Consent-First Access: Require all systems to pair digital ID authentication with real-time patient consent capture, ensuring data exchange is transparent and auditable.

- Layered Access Controls: Support credentialing tiers (e.g., IAL2/AAL2 for sensitive data, IAL1 for non-clinical services) to allow for appropriate risk mitigation without unnecessarily restricting access to health apps and services.
- Protect Proxy Relationships: Enable digital identity credentials to manage caregiver and proxy access in a secure and verifiable manner, thereby improving access for vulnerable populations while maintaining control.

CMS can accelerate the adoption of secure, reusable digital identities by providing technical integration support, value-based incentives, and safeguards for patient trust. Doing so would reduce friction in patient access, improve data security, and enable a truly interoperable health ecosystem.

PR-10. Digital Identity Credentials in Provider Workflows

a. What are the challenges and benefits for providers?

Challenges:

- Integration complexity with legacy EHR and portal systems.
- Cost and resource constraints for small and rural practices to implement CSP support.
- Patient onboarding friction, particularly for older or low-literacy populations.
- Lack of staff training and standardization across vendors.

Benefits:

- Improved authentication and auditability across data exchange workflows.
- Streamlined proxy and caregiver access through verified delegation models.
- Reduced support burden (e.g., fewer password reset requests, standardized login).
- Enhanced compliance with TEFCA and HIPAA audit traceability requirements.

b. How would requiring their use improve access to health information?

Mandating CSP-based credentials would:

- Enable patient-authorized queries across providers and QHINs without fragmented login systems.
- Unify access to longitudinal health records by verifying patient identity in a nationally trusted manner.
- Facilitate real-time cross-platform data flow, with consistent identity validation for seamless access control.
- Support scalable and secure integration with digital apps, enhancing patient engagement and interoperability.

c. What are the potential downsides?

- Equity concerns for populations without access to smartphones, stable internet, or identification documentation.
- Privacy sensitivities are particularly relevant in communities with a history of mistrust of centralized identity systems.
- Vendor resistance to changes in login systems due to contractual or legacy dependencies.
- Risk of CSP service downtime or breaches impacting large segments of access uniformly.

d. What impact would mandatory credentials have on a nationwide provider directory?

Mandating identity credentials would:

- Strengthen trust in provider directory queries by authenticating who accesses and updates data.
- Prevent directory spoofing or duplication by tying entries to verified identities.
- Enable secure, authenticated referrals and scheduling workflows to improve handoffs and patient experience.

e. How could digital identity implementation improve provider data flow?

- Providers could automatically authorize access to patient records across QHINs based on verified patient identity.
- Consent management and data segmentation would become more enforceable and scalable.
- It would reduce the administrative burden around document requests and faxed authorizations.
- Real-time eligibility, coverage, and care coordination could be tied to authenticated identities rather than ad hoc lookups.

f. Would combining FHIR addresses and identity improve data flow?

Yes. Combining FHIR endpoint addresses with CSP-backed digital identity would:

- Allow authenticated, traceable transactions across networks.
- Enable automated, policy-based routing of patient data to authorized providers.
- Support secure app launch and data integration under SMART on FHIR, reducing manual matching and increasing confidence in the data exchanged.

CMS should incentivize and gradually require the adoption of CSP-based digital identity for both patients and providers. Doing so would dramatically enhance security, interoperability, usability, and trust, especially when paired with standardized FHIR-based infrastructure.

PR-11. Role of Trust Communities in Reducing Burden and Enhancing Identity Management

Members of TEFCA trust communities, including QHINs, participants, and sub-participants, are uniquely positioned to streamline access, reduce administrative burden, and enhance data security by operationalizing a shared identity and consent infrastructure.

Reducing Provider and Patient Burden

- Federated Identity Integration: QHINs should support federated login using NIST 800-63-3 IAL2-compliant CSPs (e.g., Login.gov, ID.me), enabling providers and patients to use a single credential across the entire national network rather than managing logins for each system.
- Reusable Authorization Tokens: Trust community members can issue and honor standardized, reusable patient authorization tokens, reducing repetitive consent requests and enabling real-time data exchange across providers.
- Proxy and Caregiver Support: Identity systems within QHINs should support verified delegation workflows, enabling caregivers, legal guardians, and family members to act on behalf of patients securely, thereby improving care coordination without burdening providers with manual processes.
- Pre-Integrated Credential Flows: QHINs can embed CSP flows into their patient portals and provider platforms, making identity verification part of routine intake and access workflows, rather than a separate technical hurdle.

Enhancing Identity Management and Security

- Audit-Ready Access Controls: By standardizing authentication across the ecosystem, trust communities enable robust auditing and traceability of data access, supporting HIPAA compliance and minimizing unauthorized disclosures.
- Fraud Prevention: The use of strong digital credentials across QHINs mitigates identity spoofing, reduces the potential for misattribution of health data, and enables accurate record linking across institutions, which is essential for providing longitudinal care.
- Inter-QHIN Trust Policies: Uniform policies on identity assurance and authorization would enable seamless inter-QHIN data sharing, eliminating the need for complex re-verification and unlocking access for both patients and providers.

Strategic Recommendations

- CMS should incentivize QHIN-led CSP enrollment as part of the TEFCA onboarding process.
- ASTP/ONC should require QHINs to offer identity federation services to smaller participants lacking identity infrastructure.
- Trust community members should coordinate on identity governance frameworks to ensure consistency in how identity is used for patient access, provider access, and data exchange decisions.

TEFCA trust communities should evolve into identity and consent infrastructure providers, not just data routers. By embedding secure, interoperable digital identity services into their networks,

QHINs and participants can reduce friction, enhance trust, and support the scale needed for truly nationwide health data exchange.

4. INFORMATION BLOCKING

PR-12. Should ASTP/ONC consider removing or revising any of the information blocking exceptions or conditions within the exceptions (45 CFR part 171, subparts B through D) to further the access, exchange, and use of electronic health information (EHI) and to promote market competition?

Yes, ASTP/ONC should revise and narrow several of the existing information blocking exceptions to better fulfill the intent of the 21st Century Cures Act, promoting unfettered access, exchange, and use of electronic health information (EHI), and preventing anti-competitive behavior.

Key Revisions and Rationale:

1. Tighten the "Infeasibility Exception"

- Current problem: This exception is often invoked with vague justifications such as "technical limitations" or "burden," without standardized proof.
- Recommendation: Require documented, time-limited justification, including objective technical evidence and a remediation plan. Establish a public reporting requirement for uses of this exception to discourage abuse.

2. Refine the "Security Exception"

- Current problem: Some entities use generalized security concerns to avoid enabling API access or TEFCA connectivity.
- Recommendation: Require alignment with recognized security standards (e.g., NIST 800-53, HITRUST) and define specific, allowable conditions under which this exception may apply. ONC should also create a pre-certified security API framework to minimize ambiguity.

3. Clarify the "Content and Manner Exception"

- Current problem: Some EHR vendors and data holders use "preferred formats" as a reason to delay or deny data sharing.
- Recommendation: Require the use of standardized FHIR and TEFCA-compliant formats as the default for fulfilling requests. Allow alternatives only if both parties agree, and not as a means to unilaterally restrict access.

4. Introduce a New "Refusal to Participate in TEFCA" Clause as Information Blocking

- Proposal: Refusing to onboard to TEFCA without legitimate technical, legal, or financial justification should constitute a form of constructive information blocking.

- Entities that actively choose to remain outside of national exchange frameworks should face public accountability or disincentives if their decision materially limits access to the data they hold.

Promoting Market Competition

- Why it matters: Loosely enforced exceptions enable dominant vendors and networks to control access, resulting in vendor lock-in and undermining patient choice, provider efficiency, and digital health innovation.
- Impact of tightening exceptions: Promotes fair access to data for emerging health IT vendors, ensures QHIN-level interoperability is effective, and enables patients and providers to use best-of-breed tools rather than being confined to proprietary systems.

ONC should narrow, standardize, and enforce the use of information blocking exceptions to ensure that they are used only as intended: in good faith, under narrow circumstances. This will align market behavior with national policy goals, accelerate TEFCA adoption, and open the ecosystem to broader innovation and equitable access.

PR-13. For any category of healthcare provider (as defined in 42 U.S.C. 300jj(3)), without a current information blocking disincentive established by CMS, what would be the most effective disincentive for that category of provider?

PR-13. Most Effective Disincentives for Providers Without Current Penalties

For healthcare provider categories not currently subject to CMS-imposed information blocking disincentives, the most effective strategies would combine financial, reputational, and operational levers to drive compliance while aligning with broader value-based care goals.

Recommended Disincentives:

1. Tie Participation in CMS Payment Programs to TEFCA and Information Sharing Compliance

- Example: Make participation in Medicare programs, such as MIPS, ACO REACH, and MSSP, contingent on documented interoperability compliance, including TEFCA connectivity and API support.
- Rationale: Providers are highly motivated by participation in these programs. Making information exchange a gatekeeping criterion increases compliance without introducing new administrative complexity.

2. Withhold or Reduce CMS Incentive Payments

- Example: Impose a percentage reduction in performance-based payments for providers who fail to comply with data sharing obligations or who are flagged for repeated information blocking violations.
- Precedent: Similar approaches have been employed in Meaningful Use and eCQM reporting penalty cases.

3. Publicly Report Non-Compliant Entities

- Create a CMS-maintained "Interoperability Transparency Scorecard" that identifies providers, health systems, and vendors who are non-compliant with interoperability regulations or refuse TEFCA onboarding.
- Rationale: Public pressure and reputational risk are powerful motivators, particularly for large institutions and academic medical centers.

4. Require Interoperability as a Condition for Prior Authorization and Claims Processing

- Example: Delay or deny claims submission for organizations that do not respond to API-based data access requests or TEFCA-based record queries.
- Impact: Introduces a real-time operational consequence for failing to enable appropriate access to patient data.

5. Require Corrective Action Plans and Third-Party Audits

- **Mandate organizations flagged for information blocking to submit a CAP** and undergo an audit by a CMS-approved third party.
- Non-compliance with CAPs could escalate into exclusion from participation in specific CMS initiatives.

To ensure fairness, effectiveness, and minimal disruption, CMS should adopt a tiered model that utilizes financial and reputational disincentives to promote voluntary compliance, escalating to program exclusion and operational penalties for repeat offenders. Embedding these expectations in existing frameworks (e.g., MIPS scoring, MSSP eligibility) ensures alignment with provider priorities while reinforcing the principle that access to patient data is a clinical obligation, not a negotiable business strategy.

PR-14. How can CMS encourage providers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

Recommendations for CMS:

1. Embed Reporting into Certified EHR Workflows

CMS should work with ONC and EHR vendors to:

- Embed a "Report Information Blocking" button into certified EHR systems and provider portals.
- Enable context-aware reporting, allowing providers to file a complaint directly from the workflow where the issue occurred (e.g., a failed data request or TEFCA query denial).

This reduces friction, improves reporting accuracy, and boosts participation.

2. Make Reporting Part of TEFCA and MIPS Compliance

- Require providers participating in MIPS, ACO REACH, or MSSP to acknowledge quarterly review of interoperability failures and file complaints when appropriate.
- Include reporting frequency and follow-through as metrics or scoring components in interoperability-related quality reporting.

3. Protect Against Retaliation and Ensure Anonymity

- Allow confidential or anonymous submissions with whistleblower protections.
- Clarify that CMS/ONC will not disclose the complainant's identity to potential violators without consent, unless required by law.

4. Provide Incentives and Feedback

- Offer feedback reports or summaries to reporters, showing complaint resolution or trends.
- Consider a bonus MIPS point or recognition for providers who proactively report and document high-value interoperability challenges.

Positive Outcomes:

- Increased reporting will surface previously invisible barriers to data sharing.
- Helps target enforcement efforts toward repeat offenders, large EHR vendors, or networks that employ indirect blocking practices.
- Builds accountability and creates a feedback loop for improving TEFCA and interoperability policies.

Risks & Mitigations:

- A surge in low-quality or duplicate reports could strain ONC resources, which can be mitigated by utilizing structured reporting templates and implementing effective triage workflows.
- Some organizations may fear that reporting could sour vendor relationships, which can be addressed through strong anti-retaliation provisions and aggregate, anonymized complaint analysis.

CMS can significantly improve the utility of the Information Blocking Portal by making it easily accessible, workflow-integrated, and linked to broader incentive programs. Increased reporting will strengthen enforcement, expose systemic friction points, and ultimately accelerate adoption of TEFCA and open APIs, creating a more transparent, competitive, and patient-centered data exchange ecosystem.

D. PAYERS (PA)

PA-1. What policy or technical limitations do you see in TEFCA? What changes would you suggest to address those limitations? To what degree do you expect these limitations to hinder participation in TEFCA?

TEFCA participation must be required for all CMS-contracted payers. Currently, payer engagement with TEFCA is limited and largely voluntary, resulting in fragmented data exchange and barriers to timely care coordination. Many payers have adopted internal or vendor-specific solutions for interoperability, but these do not achieve the scalability and standardization that TEFCA offers. CMS has a unique opportunity to lead the industry by requiring Medicare Advantage, Medicaid Managed Care Organizations (MCOs), and Qualified Health Plans (QHPs) to connect and exchange data through TEFCA. Policy levers such as tying TEFCA participation to Star Ratings, quality bonus payments, and network adequacy reviews would create strong incentives. Additionally, CMS can use regulatory rulemaking to condition participation in Medicare or Medicaid programs on demonstrated TEFCA data exchange capabilities, ensuring that payer systems are no longer data silos but fully integrated partners in national health information exchange.

A primary policy limitation of TEFCA is its current voluntary participation model, particularly among payers. Without CMS-mandated participation, many Medicare Advantage (MA), Medicaid Managed Care Organizations (MCOs), and Qualified Health Plans (QHPs) are opting out or delaying onboarding. This severely limits TEFCA's value as a comprehensive, nationwide data-sharing backbone. Additionally, the technical requirements for integration remain ambiguous for many payer systems, which often operate on legacy infrastructure that is not optimized for TEFCA exchange workflows.

Key limitations include:

- Lack of payer mandate: Without regulatory enforcement, commercial and government-aligned payers face little incentive to participate.
- Limited integration standards: Payers lack clear technical guidance on how to connect to QHINs, implement FHIR-based workflows, or respond to patient-initiated queries.
- No alignment with payer-facing programs: TEFCA is not yet integrated into payer quality programs, such as Stars Ratings or network adequacy reviews.

Recommended changes:

1. Mandate TEFCA participation for all CMS-contracted payers as a condition of program participation.
2. Tie TEFCA connectivity to Star Ratings, risk adjustment validation, and quality bonus payments.
3. Publish clear payer-specific implementation guides that align with FHIR APIs, patient access requirements, and real-time event feeds.
4. Fund onboarding assistance for smaller payers and regional plans to reduce technical adoption barriers.

Impact on participation:

Without these changes, payer participation in TEFCA will remain fragmented, undermining the ability of QHINs to deliver longitudinal patient records and reducing the network's utility for both patients and providers. Mandating payer inclusion is crucial to achieving equitable and scalable interoperability across the ecosystem.

PA-2. How can CMS encourage payers to accelerate the implementation and utilization of APIs for patients, providers, and other payers, similar to the Blue Button 2.0 and Data at the Point of Care APIs released by CMS?

CMS can accelerate API adoption among payers by directly tying API functionality to core performance and compliance incentives, particularly within Medicare Advantage, Medicaid Managed Care, and Qualified Health Plan (QHP) programs. A powerful lever would be to link FHIR API implementation and utilization to Star Ratings, HEDIS quality metrics, and network adequacy evaluations. For example, higher Star Ratings could be awarded to payers who support real-time FHIR APIs for patient access, provider queries, and payer-to-payer data exchange.

To build on the success of Blue Button 2.0 and the Data at the Point of Care initiative, CMS should:

1. Mandate FHIR R4 compliance for all CMS-contracted payers across patient access, provider access, and payer-to-payer APIs.
2. Publicly benchmark API performance (e.g., latency, uptime, coverage) and highlight top-performing payers.
3. Provide technical support to small or regional payers for implementing scalable API infrastructure.
4. Establish a centralized CMS test harness and developer sandbox to ensure implementation fidelity and encourage innovation.
5. Require API endpoints to be registered and discoverable through a national directory, enabling third-party apps and providers to access services easily.

By integrating API maturity into the payer oversight and reward structure, CMS can ensure that API infrastructure becomes a foundational component of the digital health ecosystem, benefiting patients, providers, and care coordination across networks.

PA-3. How can CMS encourage payers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins?

CMS can encourage payers to adopt standardized digital identity credentials by requiring the use of NIST 800-63-3 IAL2/AAL2-compliant Credential Service Providers (CSPs)—such as Login.gov, ID.me, or CLEAR—as a condition for participation in CMS programs, including Medicare Advantage and Medicaid Managed Care.

Most payers still rely on proprietary username/password systems that are difficult for patients to manage across multiple portals. This fragmented approach creates barriers to data access, limits interoperability, and undermines patient engagement.

To drive adoption, CMS should:

1. Mandate CSP support in all beneficiary-facing portals, APIs, and third-party applications certified for use under CMS contracts.
2. Include CSP integration requirements in interoperability rules and payer certification checklists.
3. Offer implementation grants and onboarding support to smaller and regional payers to reduce the cost and complexity of CSP adoption.
4. Embed federated identity credentials into Medicare and Medicaid enrollment workflows to streamline patient setup from the outset.
5. Incentivize payers through Star Ratings or performance bonuses for demonstrating widespread CSP usage and patient uptake.

By promoting standardized digital identity through approved CSPs, CMS can reduce login fatigue, improve security, and ensure patients can authorize access to their health data across the payer ecosystem with a single, trusted credential.

PA-4. What would be the value to payers of a nationwide provider directory that included FHIR endpoints and used digital identity credentials?

National provider directories must be FHIR-based and auto-updated. A provider directory that includes FHIR endpoints and leverages digital identity credentials would deliver significant value to payers by streamlining network management, enhancing interoperability, and enabling more secure and efficient data exchange.

Key benefits include:

1. Streamlined interoperability: Payers could instantly identify and connect with provider systems using standardized FHIR endpoints, accelerating data exchange for prior authorizations, care coordination, and quality reporting.
2. Improved provider matching and referral routing: A real-time, validated directory would reduce administrative overhead by ensuring that provider data is accurate, up-to-date, and securely linked to verified digital identities.
3. Enhanced fraud prevention and compliance: Tying directory listings to digital identity credentials (e.g., NIST 800-63-3 CSPs) ensures that only authenticated, credentialed providers appear in network directories, reducing the risk of billing fraud and supporting network adequacy audits.
4. Reduced network maintenance burden: Auto-updating directories eliminate the need for manual updates across plan portals and systems, decreasing administrative costs and ensuring timely information for members and third-party apps.
5. Improved patient experience and digital navigation: With FHIR endpoints and verified identity built in, payers can support consumer tools that direct patients to high-quality, in-network providers with minimal friction and more reliable provider information.

CMS should prioritize the development of this infrastructure as a backbone for secure, standards-based provider discovery across all payers and programs.

PA-5. What are ways payers can help with simplifying clinical quality data responsibilities of providers?

Quality reporting can be simplified using Bulk FHIR and shared registries. Payers can significantly reduce the quality reporting burden on providers by leveraging Bulk FHIR APIs to extract structured clinical data directly from Electronic Health Records (EHRs) and performing the calculations themselves. This shift removes the need for provider-side computation, reduces variability in results, and enhances consistency across payers.

a. **How interested are payers and providers in EHR technology advances that enable bulk extraction of clinical quality data from EHRs to payers to allow them to do the calculations instead of the provider-side technology?**

   There is growing interest among both payers and providers in EHR advances that enable bulk extraction of clinical data. Providers increasingly favor delegated reporting models, especially when working with value-based contracts, because they offload administrative overhead while maintaining data accuracy. Payers benefit by gaining more direct insight into care quality and outcomes, enabling better performance tracking and incentive alignment.

b. **In what ways can the interoperability and quality reporting responsibilities of providers to both CMS and other payers be consolidated so investments can be dually purposed? Are there technologies that payers might leverage that would support access to real-time quality data for healthcare providers to inform clinical care, in addition to simplifying reporting processes?**

   Interoperability and reporting obligations can be consolidated by standardizing data formats and requiring participation in shared digital quality registries. These registries, built on Bulk FHIR and TEFCA infrastructure, can serve both CMS and commercial payer requirements, enabling a single pipeline for data collection, analysis, and reporting.

Payers can further support providers by:

- Offering real-time feedback dashboards on quality performance
- Embedding FHIR-based APIs into provider workflows to surface gaps in care
- Providing centralized tooling for eCQM compliance, attribution tracking, and SDOH integration

This dual-use approach allows provider IT investments to support both compliance and clinical decision-making, closing the loop between measurement and care delivery. By taking a more proactive, data-driven role, payers can become valuable partners in simplifying and scaling quality reporting.

PA-6. *<PA-6 WAS OMITTED IN THE RFI>*

PA-7. How can CMS encourage payers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

MA plans must monitor and report instances of information blocking. CMS can encourage payers to submit information blocking complaints by integrating complaint workflows into routine reporting mechanisms such as interoperability attestations, prior authorization audits, and network adequacy submissions. Embedding direct links to ONC's Information Blocking Portal within CMS reporting tools, payer dashboards, and regulatory correspondence would streamline the process and increase participation.

Additionally, CMS should:

- Educate payer compliance teams about their rights and responsibilities under the 21st Century Cures Act regarding the reporting of information blocking.
- Incorporate complaint submission metrics into Star Ratings or compliance assessments.
- Offer legal or technical support to smaller payers concerned about retaliation or unclear criteria for reporting.

Impact: Increased complaint reporting would not hinder data exchange—it would improve it. By surfacing systemic barriers (e.g., EHR vendor restrictions, unresponsive providers, or failure to comply with FHIR API access), CMS and ONC could better enforce equitable, standards-based exchange across the ecosystem. This would level the playing field, especially for smaller payers or provider groups, and reinforce TEFCA participation as a shared obligation.

Over time, a transparent and enforced complaint mechanism would accelerate adoption of interoperable systems, protect patient access rights, and strengthen trust in CMS-led interoperability programs.

TECHNOLOGY DEVELOPERS (TD)

E. Technology Vendors, Data Providers, and Networks

1. Ecosystem

TD-1. What short-term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

CMS should launch an interoperability sandbox and require QHIN access for apps. The sandbox would function as a secure, simulated environment where developers can test applications against real-world health data standards, such as FHIR, TEFCA query protocols, and SMART on FHIR authorization. Participants would include certified technology vendors, early-stage health IT startups, Qualified Health Information Networks (QHINs), and payer organizations seeking to

validate API compliance and system compatibility. Expected outcomes include faster time-to-market for compliant apps, improved security practices through monitored scenarios, and broader adoption of standardized APIs across digital health tools. The sandbox should consist of a public scoring dashboard to evaluate use-case readiness and TEFCA alignment, with priority given to tools that serve underserved populations and advance equity goals, and that require Qualified Health Information Network (QHIN) access for their applications.

TD-2. Regarding CMS Data, to stimulate developer interest:

a. What additional data would be most valuable if made available through CMS APIs?

Developers would benefit most from access to real-time lab results, imaging reports, admit-discharge-transfer (ADT) notifications, prior authorization status, and coverage eligibility data through standardized CMS APIs. These data types are essential for powering clinical decision support, care coordination, and patient engagement applications.

b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API?

The most valuable data sources to complement Blue Button 2.0 include:

- Clinical data via TEFCA (EHR-sourced data through QHINs)
- Social determinants of health (SDOH) data from community organizations
- Home health and hospice records
- Imaging and pathology data
- Medicaid encounter data

These sources provide a more comprehensive, longitudinal view of patient health, enabling the precision targeting of interventions.

c. What obstacles prevent accessing these data sources today?

Barriers include:

- Lack of payer and provider TEFCA participation
- Siloed data systems with no FHIR API support
- Inconsistent consent and authorization frameworks
- Limited infrastructure for real-time query and event-driven data exchange
- Proprietary data contracts or non-standard API implementations

d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

CMS should introduce and certify support for the following API types:

- FHIR-based Prior Authorization APIs (per the Da Vinci standard)

- Real-time ADT and care transition event feeds
- Provider directory APIs with verified FHIR endpoints and CSP-backed identity
- Payer-to-payer data exchange APIs for longitudinal record continuity
- Bulk FHIR APIs for population health and quality measurement
- SDOH referral and outcome tracking APIs

By expanding CMS API offerings and aligning them with TEFCA exchange, CMS can dramatically increase developer engagement and unlock innovation that benefits patients, providers, and payers alike.

2. Digital Identity

TD-3. Regarding digital identity implementation:

a. What are the challenges and benefits?

Challenges:

- Integration complexity: Many legacy EHR systems and payer portals are not yet configured to support federated digital identity workflows or NIST 800-63-3 CSP protocols, requiring significant upgrades.
- User onboarding friction: Some patient populations, particularly the elderly, underserved, or those with limited English proficiency, may encounter barriers during digital credential enrollment due to access to technology or literacy gaps.
- Cost concerns: Smaller technology developers or safety-net organizations may face licensing or development costs to integrate certified identity verification solutions.
- Lack of standardization across systems: Even when CSPs are supported, inconsistent implementation across providers and payers creates fragmented identity experiences.

Benefits:

- Streamlined and secure access: Patients and providers can access their health data across multiple systems using a single, verified credential, reducing login fatigue and improving usability.
- Enhanced consent management: Digital identity facilitates more granular, auditable consent and delegation (e.g., proxy access), thereby supporting both compliance and patient engagement.
- Reduced fraud and impersonation: Strong identity assurance lowers the risk of unauthorized access, particularly in high-stakes contexts such as prior authorization, record sharing, and benefit enrollment.
- Interoperability enabler: Digital identity is foundational to seamless TEFCA-based exchange and patient-matching across platforms.

b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?

Requiring NIST 800-63-3 IAL2/AAL2 digital identity credentials would strengthen cybersecurity across the healthcare ecosystem by enforcing multi-factor authentication and verified identity before granting access to sensitive clinical and financial information. It would also:

- Mitigate identity spoofing across QHINs and third-party applications.
- Facilitate real-time, cross-platform access to patient-authorized records under TEFCA.
- Enable interoperable credentialing for both providers and patients, streamlining onboarding and access control across systems.

This would also create a standard trust layer between apps, APIs, and exchange participants, driving adoption of patient- and provider-directed data exchange at scale.

c. What impact would mandatory use of the OpenID Connect identity protocol have?

Requiring OpenID Connect (OIDC) as the identity protocol standard would:

- Ensure compatibility across applications, reducing vendor-specific identity silos and accelerating developer adoption.
- Support delegated access models, such as caregivers, family proxies, and authorized apps acting on behalf of patients.
- Facilitate secure session management and token-based authorization for dynamic API interactions, such as SMART on FHIR apps.
- Enable federation across TEFCA participants, allowing a single identity credential to function across multiple networks, such as QHINs, payers, and providers.

This would lower the barrier for developers building identity-aware applications, reduce the complexity of implementing secure access, and establish a scalable digital identity framework for the healthcare ecosystem.

3. Technical Standards and Certification

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

Only certified open APIs should qualify for CMS programs. CMS can better encourage the use of open, standards-based, publicly available APIs by mandating their use in all accredited health IT products and federal health programs. This can be supported through the expansion of the ONC Health IT Certification Program to require adherence to FHIR-based APIs and public endpoint directories. Additionally, CMS can incentivize compliance by linking payment eligibility, program participation, and quality bonuses to the demonstrated use of non-proprietary Application Programming Interfaces (APIs). Public transparency dashboards showing API performance and adoption rates across vendors can further foster competition and accountability. Lastly, CMS could offer innovation grants or technical assistance for developers building open APIs, particularly those addressing underserved populations and interoperability challenges.

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

CMS must maintain a FHIR directory with verified endpoints. A nationwide directory of FHIR endpoints would serve as a critical utility for simplifying and scaling access to health information across patients, providers, and payers. For patients, it enables the use of digital tools to access and connect with their health records, regardless of the care setting or health information technology vendor. For providers, it streamlines referrals, data queries, and inter-organizational communication by automatically identifying endpoint connections and trust relationships. For payers, it supports efficient data exchange for claims processing, prior authorizations, and quality measurement.

This directory should be published and maintained by the CMS or a designated federal entity, such as the Office of the National Coordinator for Health Information Technology (ONC/ASTP), to ensure neutrality, regulatory authority, and long-term sustainability. To ensure accuracy, endpoint registration should be tied to certification processes or TEFCA QHIN onboarding. Users should not bear a cost to access or query this directory; the value of such infrastructure lies in its openness and utility across the entire ecosystem. CMS can fund its maintenance through certification or data access programs, ensuring equitable participation and broad adoption.

TD-6. What unique interoperability functions does TEFCA perform?

TEFCA should replace fragmented network models. Its unique interoperability functions include a standardized legal and operational framework for data exchange, a common agreement for participant responsibilities, and a designated Qualified Health Information Network (QHIN) structure that establishes trusted, nationwide data sharing. TEFCA supports both query-based and push-based data exchange across disparate systems and geographies, providing a single, unified mechanism to access longitudinal health records, regardless of location or vendor. Unlike private networks or regional HIEs, TEFCA is designed to ensure that any participant connected through a QHIN can seamlessly retrieve clinical, administrative, and social health data from any other participant, provided that appropriate authorization is in place. This governance-layered, federated exchange model is scalable, inclusive, and designed to resolve long-standing challenges in nationwide interoperability.

a. **What existing alternatives should be considered?** Existing options to TEFCA include point-to-point connections, proprietary payer-provider data exchange networks, and regional health information exchanges (HIEs). While these solutions have some value locally or within specific verticals, they often lack standardization, portability, and comprehensive governance.
b. **Are there redundant standards, protocols or channels or both that should be consolidated?** Redundant standards and protocols such as Direct Messaging, custom API formats, and legacy HL7 v2 transport should be consolidated or deprecated. Consolidation around FHIR R4 and TEFCA's QHIN-based model would streamline interoperability and lower integration burdens across the ecosystem.

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

USCDI has significantly advanced interoperability by creating a core set of standardized data elements that are universally expected to be exchanged between health IT systems. However, while its foundation is strong, it has limitations in supporting the full scope of clinical and administrative data required for holistic care delivery and advanced analytics.

a. **Does it contain the full extent of data elements?** No, USCDI does not yet include all the data elements required for comprehensive interoperability. Crucial gaps remain, including genomic data, patient-reported outcomes, device-generated health data, and detailed social determinants of health (SDOH).
b. **If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?** These gaps are partly due to the scope limitations of the current USCDI definition, which focuses primarily on traditional clinical data categories. Additionally, the implementation of USCDI varies across vendors, resulting in reduced consistency.
c. **If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?** Adding more data elements to USCDI adds value by enabling richer clinical insights and better care coordination. However, expanding USCDI must be accompanied by robust guidance on implementation to avoid creating scoping or compliance burdens. One approach would be to adopt a tiered model, with basic USCDI for compliance and extended USCDI+ for advanced use cases.
d. **Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver**? Given advances in AI and natural language processing, many stakeholders may prefer a less structured but broader format that enhances data comprehensiveness. While this would require more processing on the part of recipients, it could increase access to valuable contextual information. CMS should consider providing parallel support for both structured and semi-structured data flows, allowing for flexibility and continued innovation while maintaining a standards-based core.

TD-8. What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

The most effective certification criteria under the ONC Health IT Certification Program are those that enforce modular, standards-based interoperability. Specifically, requirements for certified API support using HL7 FHIR, including support for SMART on FHIR for authorization and authentication, have proven instrumental in facilitating standardized access to data. Certification should also ensure implementation of Clinical Decision Support (CDS) Hooks, Bulk FHIR for population-level data, and FHIR Subscriptions for real-time updates. These tools establish the foundation for event-driven care coordination, predictive analytics, and the assembly of longitudinal patient records. To further improve certification efficacy, CMS and ONC should

introduce active surveillance of API uptime, data completeness, and response quality to validate the real-world performance of certified modules.

TD-9. Regarding certification of health IT:

APIs should be prioritized over UI features. Coverage must include unstructured data to ensure that critical insights locked in scanned records or narrative text are available to downstream tools and users.

a. **What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?** Redefining certification to prioritize API-enabled capabilities would encourage developers to design modular, scalable systems that emphasize data liquidity. It promotes innovation by enabling third-party apps to access the same data as the native Electronic Health Record (EHR).
b. **What would be the drawbacks?** The primary drawback is that focusing solely on APIs may deprioritize core usability and safety functions in the user interface layer, which are still essential for clinician workflow.
c. **How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?** ONC could revise health IT certification to require APIs capable of exchanging the full EHI, including scanned documents and free text, by implementing FHIR DocumentReference and supporting C-CDA to FHIR mapping. Certification could require support for note type and attachment metadata.
d. **What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data**? CMS could tie reimbursement bonuses or quality score improvements to providers' responsiveness to API-based requests, measured by completeness, latency, and semantic accuracy. Public dashboards could display provider API performance as a reputational incentive.
e. **How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?** EHRs with Bulk FHIR support could automate quality reporting by pre-generating performance datasets, reducing the manual burden. To demonstrate its benefits, the system must support risk adjustment variables and stratification codes. Concerns include data standardization across vendors and ensuring that providers can validate outputs before submission. CMS should fund pilot programs to evaluate the accuracy and efficiency gains from automated bulk reporting using certified Application Programming Interfaces (APIs).

TD-10. For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj-11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without

special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?

Full EHI access should be enforced unless restricted by law. To further implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj–11(c)(5)(D)(iv)), ASTP/ONC should require certified health IT developers to demonstrate real-time, API-based access to all data elements contained in a patient's EHR, including structured, semi-structured, and unstructured formats. Certification should extend beyond core clinical data to include attachments, scanned documents, and historical notes, provided they are permissible under HIPAA and applicable state privacy laws. APIs must be designed to allow third-party applications to query and consume this data without requiring "special effort", thereby eliminating the need for manual exports, unsupported data formats, and contractual limitations.

ONC should conduct random audits of certified APIs in live production environments to assess their completeness and real-world accessibility. Developers that condition access to these APIs on additional fees, proprietary licensing, or specific partner agreements should be flagged for noncompliance. Furthermore, CMS should align payment models to reward providers that expose EHI comprehensively and in a timely manner. Together, these actions will fulfill the Cures Act's goal of ensuring patients and authorized actors can access, exchange, and use health data without unnecessary technical or policy barriers.

TD-11. Routine EHI export via Bulk FHIR should be mandated. Now that EHI export has been in production since January 1, 2024, further steps are needed to strengthen its utility:

a. **Should this capability be revised to specify standardized API requirements for EHI export?** This capability should be revised to specify standardized API requirements. Using FHIR Bulk Data Export (Flat FHIR/NDJSON) as a universal format would reduce variation and improve integration with downstream systems. This standardization should extend to metadata definitions, export parameters, and authentication protocols.
b. **Are there specific workflow aspects that could be improved?** Workflow improvements should include automated export scheduling, user-configurable export scopes (e.g., by encounter type or clinical domain), and integration with patient-facing applications to support transparency and portability.
c. **Should CMS consider policy changes to support this capability's use?** CMS should consider policy changes to align quality program requirements with EHI export capabilities. For example, CMS could require that all providers participating in Medicare share a quarterly EHI export with designated registries or Qualified Health Information Networks (QHINs). Additionally, payment incentives could reward organizations that make export data available to patients in real time through third-party apps.

4. Data Exchange

TD-12. Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

CMS should endorse TEFCA-aligned third-party data providers. Endorsement should be based on clear, measurable criteria such as adherence to HL7 FHIR standards, active QHIN participation, auditability of access controls, and documented data provenance. These organizations should demonstrate that they meet or exceed the same interoperability and privacy safeguards as CMS-certified entities. Endorsing such networks would help scale innovation by giving providers and developers broader access to validated data sources.

CMS should consider requiring transparency in the business practices of endorsed networks, particularly regarding pricing, data use agreements, and patient consent models. Metrics for endorsement could include uptime reliability, data completeness, real-time accessibility, and support for patient-centered data exchange. To ensure equity and scalability, access to endorsed networks should be cost-neutral for providers and app developers serving beneficiaries of Medicare and Medicaid. Publicly listing endorsed entities would improve trust and accelerate the ecosystem's migration toward TEFCA-aligned interoperability.

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

APIs must support both USCDI and complete EHI. Providing full API-based access to a patient's entire electronic health information (EHI) unlocks opportunities for more comprehensive, longitudinal care, precision medicine, and population health insights. Access to clinical notes, scanned documents, attachments, and observational data elements allows providers and developers to deliver more nuanced risk stratification, proactive care coordination, and targeted patient engagement.

a. **What are the primary obstacles to this?** The primary challenges to achieving full EHI access via APIs include inconsistent data modeling across EHR vendors, data access fees, and the lack of certified mechanisms to expose semi-structured and unstructured content. Many EHR systems still require manual chart pulls for older data, scanned faxes, and historical narrative notes.
b. **What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?** The tradeoffs between USCDI and full EHI lie in the balance between standardization and comprehensiveness. USCDI offers high consistency and interoperability, but a limited view. Full EHI is richer but harder to process, requiring more robust data normalization and interpretation.

Given today's advances in AI and natural language processing, many organizations are capable of ingesting broader, less-structured data and extracting clinical value. CMS should support this by requiring APIs to expose both USCDI and complete EHI with appropriate metadata, while investing in tools to help recipients derive structured insights from complex formats. This dual approach strikes a balance between innovation and interoperability.

TD-14. Regarding networks' use of FHIR APIs:

a. How many endpoints is your network connected to for patient data sharing? What types, categories, geographies of endpoints do you cover? Are they searchable by National Provider Identifier (NPI) or organizational ID?

beHuman's platform, built on top of Health Gorilla's national QHIN infrastructure, connects to over 100,000 FHIR endpoints spanning all 50 states. These include:

- Hospitals and health systems using certified EHRs (e.g., Epic, Cerner, Meditech)
- Independent practices and urgent care centers
- National and regional labs (e.g., LabCorp, Quest, BioReference)
- Pharmacies and imaging centers
- FQHCs and community-based providers

All endpoints are indexed and searchable by National Provider Identifier (NPI), Taxpayer Identification Number (TIN), and organizational ID, enabling automated routing and filtering during patient matching, record retrieval, and referrals.

b. How are these connections established (for example, FHIR (g)(10) endpoints, TEFCA/Integrating the Health Enterprise (IHE) XCA, or proprietary APIs)?

Connections are established through a hybrid model, including:

- FHIR (g)(10) endpoints for certified EHR data access under ONC interoperability rules.
- IHE XCA/XDS.b protocols are used in TEFCA-compliant exchange between QHINs.
- Direct integration via proprietary APIs, where TEFCA or FHIR endpoints are unavailable, but access is contractually permitted.
- Bulk FHIR and SMART on FHIR apps for population-level access and provider-initiated exchange.

This approach maximizes connectivity while ensuring compliance with national interoperability and privacy standards.

c. Do you interconnect with other networks? Under what frameworks (for example, TEFCA, private agreements)?

Yes, beHuman's infrastructure, via Health Gorilla's QHIN designation, is interconnected with other QHINs under the TEFCA framework, enabling cross-network data exchange with a common trust agreement and technical baseline.

In addition, we plan to maintain private interoperability agreements with major EHR vendors, national labs, and payer platforms to supplement TEFCA-based exchange where needed. These private frameworks serve as transitional bridges until full TEFCA adoption is achieved.

By combining TEFCA governance with flexible implementation models, we ensure robust, standards-aligned, and scalable access to clinical data across the nation.

TD-15. Regarding bulk FHIR APIs:

Bulk queries support analytics and VBC; CMS should require them.

a. **How would increased use of bulk FHIR improve use cases and data flow?** Increased use of bulk FHIR APIs would enhance a wide range of use cases, including population health management, predictive modeling, quality reporting, and preventive care outreach. These APIs enable the scalable and efficient retrieval of large patient cohorts, making it feasible for ACOs, health plans, and digital health companies to track clinical metrics, identify care gaps, and proactively engage with high-risk populations. They also enable near real-time integration with data lakes and analytics platforms, which can power longitudinal insights and support social risk stratification.
b. **What are the potential disadvantages of their use?** Possible disadvantages include greater demand on server resources, challenges in synchronizing incremental updates, and risks related to data governance and security if significant exports are not properly scoped or encrypted. There may also be variation in how EHR vendors implement Bulk FHIR endpoints, which could lead to inconsistent outputs across platforms. CMS should consider standardizing export parameters, encouraging the development of test harnesses, and establishing certification metrics for Bulk FHIR performance and completeness.

TD-16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?

Point-to-point does not scale; QHINs must be the foundation. Maintaining point-to-point models requires custom integrations, redundant agreements, and ongoing bilateral maintenance, resulting in unsustainable complexity and high transaction costs as the number of participants increases. In contrast, shared network infrastructure, such as TEFCA, enables a scalable and standardized exchange that reduces administrative friction and improves real-time data access across payers, providers, and developers.

a. **Do current rules encourage scalable network participation?** Current regulations do not sufficiently encourage scalable participation. Voluntary TEFCA participation means that smaller networks and vendors often lack the incentive to invest in interoperability infrastructure. CMS should mandate QHIN connectivity for Medicare and Medicaid participants to establish a national baseline standard for connectivity.
b. **What changes would improve alignment (for example, API unification, reciprocal access)?** Improved alignment could be achieved through API unification (e.g., FHIR R4 as the mandated standard), reciprocal data access mandates, and financial incentives tied to successful bidirectional data exchange. CMS could also require that all covered entities expose their endpoints and data access policies in a public TEFCA directory, fostering transparency and network fluidity.

TD-17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

CMS should publish QHIN performance benchmarks. Additionally, CMS and ONC should play an active role in ensuring the long-term viability of healthcare data-sharing networks by funding initial infrastructure development, subsidizing onboarding for smaller or safety-net providers, and providing outcome-based incentives for participation. Ensuring a sufficient supply of participating nodes (providers, payers, labs, developers) and meaningful demand (use cases tied to care coordination, payment, and public health reporting) is critical to achieving scale. CMS can also support viability by embedding TEFCA compliance into federal program eligibility, such as Medicare participation, and establishing value-based payment adjustments based on the success of the exchange. Regularly published metrics should include query volume, match rates, latency, and data completeness across QHINs, helping guide investment and build ecosystem trust.

5. Compliance

TD-18. Information blocking:

Information blocking encompasses a range of vendor and provider behaviors that restrict data access despite possessing the necessary technical capabilities.

a. **Could you, as a technology vendor, provide examples for the types of practices you have experienced that may constitute information blocking. Please include both situations of non-responsiveness as well as situations that may cause a failure or unusable response.** Examples we have encountered include health IT vendors that impose exorbitant fees to activate standard FHIR APIs, even when they are already certified under ONC programs. Others create artificial data throttling or delay the release of documentation necessary to authenticate connections. Some providers reject QHIN or TEFCA connections without justification or implement firewalls that permit only internal data flows while denying access to certified digital health tools.
b. **What additional policies could ASTP/ONC and CMS implement to further discourage healthcare providers from engaging in information blocking practices?** To further prevent these practices, CMS and ASTP/ONC should expand enforcement authority by requiring real-time API performance metrics from certified systems and enabling public complaint portals for patients and developers. Providers engaged in repeated blocking behavior should be penalized through payment adjustment or program exclusion.
c. **Are there specific categories of healthcare actors covered under the definition of information blocking in section 3022(a)(1) of the Public Health Service Act (PHSA) that lack information blocking disincentives?** Specific categories of actors, such as health information networks or affiliated hospital IT departments, may fall under the definition of information blocking but are rarely investigated. CMS should close these gaps by clarifying disincentives across all actor types and

ensuring enforcement applies uniformly, particularly to intermediaries that control routing or policy configurations.

TD-19. Regarding price transparency implementation:

CMS should require the availability of machine-readable price data.

a. Current shortcomings include incomplete files, non-standard data formats, missing negotiated rate data, and poor searchability. Many machine-readable files are large, unindexed, and hosted in formats that consumers or app developers cannot easily parse. Timeliness is also an issue, as updates are often infrequent or published retroactively.
b. Workflows that would benefit most include provider-patient shared decision-making, advanced explanation of benefits (AEOB) planning, and automated cost estimation during referral or scheduling. Payers and navigation tools could also streamline prior authorization workflows using accurate pricing.
c. For patients, improvements like structured, filterable out-of-pocket estimates by location and service category would enable informed decision-making. For providers, direct API integration with cost estimates would enhance clinical decision support. For CMS, this transparency supports the implementation of value-based care and the detection of fraud.
d. To further motivate development, CMS could launch incentive programs for certified transparency tools, require payer reporting of real-time API availability, and create validation benchmarks to ensure usability. Public ranking of payer transparency compliance could also drive competitive improvements.

VALUE-BASED CARE (VB)

VB-1. What incentives could encourage APMs such as accountable care organizations (ACOs) or participants in Medicare Shared Savings Program (MSSP) to leverage digital health management and care navigation products more often and more effectively with their patients? What are the current obstacles preventing broader digital product adoption for patients in ACOs?

Providers require real-time access to comprehensive patient data, including lab results, care plans, medication adherence updates, ADT (Admit, Discharge, Transfer) notifications, and specialist referrals, to manage populations under value-based care contracts effectively. AI-based clinical decision support (CDS) tools, such as risk stratification models, automated gap-in-care alerts, and predictors of chronic disease progression, have demonstrated improved quality scores and reduced avoidable hospitalizations. For example, predictive AI algorithms integrated with longitudinal data through TEFCA have successfully flagged patients at risk of cancer or diabetes complications in pilot programs. By combining real-time data with intelligent automation, providers can close care gaps faster and deploy interventions earlier, thereby improving outcomes and driving success in value-based payment models.

To encourage APMs such as ACOs or MSSP participants to adopt digital care navigation tools more frequently and effectively, CMS could offer enhanced shared savings for organizations that integrate certified tools into their workflows. Further incentives could include upfront infrastructure payments, streamlined compliance reporting, and performance bonus adjustments tied to verified patient engagement metrics. Current obstacles include EHR integration limitations, insufficient reimbursement mechanisms for digital engagement, and a lack of technical support for onboarding tools that serve marginalized populations. By embedding TEFCA-driven data sharing, mandating CSP-based digital identity, and reducing administrative complexity, CMS can promote broader and more equitable adoption of digital health management solutions in value-based care.

VB-2. How can key themes and technologies such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement be better integrated into APM requirements?

CMS should subsidize TEFCA connectivity and reward digital equity tools. To further advance the adoption of advanced payment models (APMs), CMS should explicitly integrate themes such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement into program requirements. These capabilities can be promoted through a digital certification pathway aligned with TEFCA, structured around API readiness, real-time data usage, and demonstrated impact on health equity. CMS should also create a scoring framework within APMs (e.g., MSSP, ACO REACH) that rewards the implementation of AI-driven care gap closure, automated risk detection, and digital-first navigation platforms. Current barriers to adoption—such as EHR integration challenges, low reimbursement for virtual care coordination, and lack of infrastructure support—can be addressed by offering supplemental per-beneficiary-per-month (PBPM) payments and technical assistance grants. TEFCA-enabled workflows should be made the default data exchange framework for all care coordination and quality measurement processes within these models.

VB-3. What are essential health IT capabilities for value-based care arrangements?

Equity dashboards utilizing TEFCA data should be incorporated into all models. Key health IT features for value-based care include real-time patient data aggregation, predictive analytics for risk stratification, digital tools for patient engagement, longitudinal records via TEFCA, and care coordination platforms that track referrals and outcomes. Other critical capabilities include care planning tools with shared decision-making features, patient event notifications triggered by ADT feeds, data extraction and normalization engines, and integrated quality performance measurement modules that align with CMS reporting formats. Robust patient attribution logic and identity matching algorithms, particularly those that support probabilistic and referential matching across systems, are also vital.

a. Additional capabilities, such as access to real-time claims data and remote monitoring integration through device APIs, enhance visibility and patient empowerment. These systems must support FHIR and C-CDA formats for ingestion and processing.

b. Health IT features that have proven especially valuable include AI-based triage support. These multi-channel outreach tools adapt to patient preferences, including SMS, email, and voice, as well as care navigation platforms that surface localized community resource data. TEFCA-backed interoperability ensures that all these capabilities are data-rich and context-aware, improving equity, efficiency, and performance in value-based programs.

VB-4. What are the essential data types needed for successful participation in value-based care arrangements?

CMS should certify tools that meet benchmarks for equity, access, and interoperability. Key data types for value-based care include laboratory results, ADT notifications, social determinants of health, imaging reports, immunizations, prescriptions, care plans, problem lists, vital signs, and functional status assessments. Claims data, including diagnosis codes, procedure codes, and payment histories, also play a crucial role in risk adjustment, attribution, and performance measurement. CMS should require TEFCA-aligned platforms to support the exchange of these data types via FHIR APIs.

a. Examples include patient-generated health data from remote monitoring devices, clinical notes, HCC coding data, and eligibility and coverage data—all of which support longitudinal care and proactive interventions.
b. Additionally, data related to transportation needs, housing status, food insecurity, and language preferences have proven especially valuable in driving improved outcomes in underserved populations. These data types enable digital tools to connect patients with specific resources and services, and should be integrated into certified digital platforms via structured Application Programming Interfaces (APIs).

2. Compliance and Certification

VB-5. In your experience, how do current certification criteria and standards incorporated into the ONC Health IT Certification Program support value-based care delivery?

Critical data, including labs, ADTs, SDOH, imaging, care plans, immunizations, and claims, must be standardized through TEFCA to enable effective value-based care. Current ONC certification criteria support value-based care by requiring baseline interoperability, but additional steps are needed. Certification should mandate real-time access to structured data elements, standardized terminology (e.g., SNOMED, LOINC), and the ability to reconcile and aggregate longitudinal data across encounters.

ONC could strengthen its criteria by requiring certified health IT products to demonstrate TEFCA QHIN integration, support for CDS Hooks, and seamless ingestion of external data into core EHR workflows. Furthermore, accredited systems should be required to expose and consume risk adjustment variables, SDOH indicators, and attribution logic in support of value-based performance measurement. By aligning these capabilities with CMS's payment models,

ONC certification can directly reinforce the goals of accountable care and population health improvement.

VB-6. What specific health information technology capabilities that could benefit APMs are not currently addressed by existing certification criteria and standards that should be included under the ONC Health IT Certification Program?

CMS should fund onboarding for FQHCs and mandate a connection to QHIN for SDOH providers. Additionally, CMS and ONC should expand the ONC Health IT Certification Program to address currently unmet needs in value-based care. Specifically, certification should encompass capabilities such as care gap detection, real-time patient risk scoring, digital care planning and coordination tools, patient attribution engines, and support for integration with community-based organizations that address social determinants of health. Certified systems should be able to ingest and act on real-time event notifications (e.g., ADTs), offer API endpoints for remote monitoring data, and integrate with national patient ID matching services. Certification criteria should further require demonstration of support for longitudinal care pathways that span clinical and non-clinical providers. These standards will ensure that APM participants—particularly those in safety-net settings—have access to interoperable, outcomes-focused tools that support coordinated, equitable care.

VB-7. How can technology requirements for APMs, established through CEHRT or other pathways, reduce complexity while preserving necessary flexibility?

Use TEFCA analytics to evaluate engagement, outcomes, and equity. Technology requirements for APMs, when aligned with TEFCA infrastructure, can reduce complexity by establishing standardized APIs, data formats, and identity protocols across all participants. To preserve necessary flexibility, CMS and ONC could allow for tiered certification paths—where core requirements are uniform, but optional modules support advanced features like AI-based triage, social risk stratification, and community resource linkage. Additionally, embedding TEFCA analytics dashboards into APM workflows enables organizations to monitor engagement rates, health outcomes, and disparities in near real time. This data-driven feedback loop simplifies compliance, reduces redundancy, and supports continuous improvement while allowing organizations to customize care pathways based on population needs.

VB-8. How can other HHS policies supplement CEHRT requirements to better optimize the use of digital health products in APMs? As an example, requirements under the Conditions of Participation for hospitals (42 CFR 482.24(d)) require hospitals to transmit electronic patient event notifications to community providers. What barriers are in place preventing APM participants from receiving the same notifications?

Tech infrastructure must be reimbursable under Value-Based Care (VBC) models. CMS should supplement CEHRT requirements with Conditions of Participation and other HHS policies that incentivize digital integration. For instance, while hospitals are currently required to transmit ADT-based patient event notifications under 42 CFR 482.24(d), many APM participants, such as Accountable Care Organizations (ACOs) and community-based providers, lack access due to several barriers. These include incomplete TEFCA connectivity, inconsistent implementation of

event notification protocols across EHR vendors, a lack of funding for integration, and the absence of enforcement for bidirectional data sharing.

To overcome this, CMS should expand notification requirements to include all APM participants and provide targeted funding for event listeners and integration capabilities. Additionally, CMS can align incentives by including real-time data exchange compliance as a scored component within value-based programs, such as ACO REACH and MSSP. ONC should establish a certification requirement for event notification APIs and standardized delivery mechanisms to ensure usability across diverse care settings. These efforts would promote equitable and coordinated care by making patient transitions visible and actionable across the entire healthcare ecosystem.

VB-9. What technology requirements should be different for APM organizations when comparing to non-APM organizations (for example, quality reporting, and interoperability)?

Technology requirements for APM organizations should reflect their heightened accountability for outcomes, coordination, and equity. Unlike non-APM organizations, APM participants require infrastructure that supports continuous population-level data exchange, longitudinal record access, and bidirectional care coordination. Interoperability requirements should go beyond basic data sharing to include TEFCA-based integration, real-time ADT notifications, and bulk FHIR access for quality and risk reporting.

In quality reporting, APMs should be equipped with automated extraction of eCQMs and the ability to generate stratified performance metrics by race, language, and geography in bulk. Certification criteria should also include support for attribution logic, risk adjustment algorithms, and care gap detection, capabilities not typically required for fee-for-service providers.

To address digital equity, certified tools should also support multilingual interfaces, offline functionality, and adaptive content delivery for low-bandwidth or low-literacy environments, ensuring that APMs serving underserved populations can meet engagement targets.

By aligning technology requirements with the advanced functions APMs must perform, CMS can ensure these organizations are positioned to succeed in delivering value-based care.

VB-10. In the Calendar Year (CY) 2024 Physician Fee Schedule final rule (88 FR 79413), CMS established that CEHRT requirements for Advanced APMs beyond those in the "Base EHR" definition should be flexible based on what is applicable to the APM that year based on the area of clinical practice. What certification criteria should CMS identify under this flexibility for specific Advanced APMs, or for Advanced APMs in general? Are there specific flexibilities or alternatives to consider for smaller or resource-constrained (such as rural) providers in meeting CEHRT requirements without compromising quality of care or availability of performance data?

CMS should reward TEFCA-powered, equity-driven digital innovation. In implementing CEHRT requirements for Advanced APMs under the flexibility established in the CY 2024

Physician Fee Schedule final rule (88 FR 79413), CMS should identify certification criteria that reflect the specific data, workflows, and population needs of each APM model. Rather than a one-size-fits-all approach, CEHRT requirements should be modular and outcomes-driven, with core capabilities required across all models and optional pathways tailored to specialty, population focus, or care delivery setting.

Recommended criteria for Advanced APMs include:

- Integration with TEFCA via QHIN connectivity for longitudinal data exchange.
- Bulk FHIR support for population-level analytics and quality measurement.
- CDS Hooks and SMART on FHIR APIs for embedding clinical decision support and patient engagement tools.
- Structured data capture for SDOH, referrals to community services, and patient-reported outcomes.
- Real-time event notifications (ADT, lab, imaging) to support care transitions and proactive intervention.

Flexibilities or alternatives for smaller or resource-constrained providers (e.g., rural or FQHCs) should include:

- Certification subsidies or grants for onboarding to TEFCA and cloud-based CEHRT platforms.
- Use of shared digital infrastructure or regional digital health collaboratives to meet interoperability requirements.
- Extended implementation timelines or streamlined certification checklists aligned with essential VBC capabilities (e.g., care gap closure, basic eCQM reporting).
- Allowing validated third-party tools, such as patient navigation platforms or mobile-first EHRs, that meet CMS data reporting standards but are not formally certified under ONC's broader CEHRT program.

By structuring CEHRT expectations around clinical relevance, interoperability, and equitable access, CMS can drive innovation and performance across a diverse range of APM participants without compromising the quality of care or accountability.

CMS must transition from optional frameworks to mandatory infrastructure to support a patient-centered, digitally enabled healthcare system. TEFCA is the most credible path to unifying health data exchange at scale. Its promise will only be realized if:

- Payers, labs, and certified technology vendors are mandated to participate.
- Provider-to-payer exchange and patient-initiated access are standardized through QHINs.
- Digital identity via credential service providers is universally implemented.

2. Technical Standards

VB-11. What specific interoperability challenges have you encountered in implementing value-based care programs?

In implementing value-based care (VBC) programs, we have encountered several specific interoperability challenges that hinder care coordination, quality measurement, and outcomes-based reimbursement:

1. **Fragmented Access to Longitudinal Patient Data**
    - VBC models require a complete view of a patient's care journey across multiple providers and settings. However, due to limited TEFCA participation by commercial payers, labs, and non-hospital providers, critical data such as lab results, prior diagnoses, and discharge summaries are often inaccessible or delayed.
2. **Inconsistent Use of Standard APIs Across EHRs**
    - Many EHR vendors do not consistently implement FHIR APIs beyond the minimum patient access requirements. This inconsistency prevents real-time integration with third-party tools for risk stratification, care gap alerts, and remote monitoring.
3. **Delayed or Missing ADT Notifications**
    - Timely Admit, Discharge, and Transfer (ADT) notifications are crucial for managing care transitions and preventing readmissions. Yet, many providers—especially those in rural or safety-net settings—do not participate in networks that offer standardized ADT feeds.
4. **Limited Integration with Community-Based and SDOH Providers**
    - Social determinants of health (SDOH) data are crucial for achieving health equity in VBC; however, interoperability between clinical systems and community-based organizations (e.g., housing, nutrition, transportation) is virtually nonexistent due to a lack of standardization and insufficient funding.
5. **Lack of Unified Patient Identity Across Systems**
    - Without a universal digital identity infrastructure, matching patients across providers, payers, and health IT systems remains error-prone, especially in underserved or mobile populations.
6. **Burden of Quality Reporting and Attribution**
    - Providers are often required to report quality metrics using proprietary formats or duplicative systems, which increases their administrative burden. Attribution logic is opaque and varies by payer, making population health management inefficient and reactive.

To overcome these challenges, CMS should mandate TEFCA connectivity for all VBC participants, require universal support for FHIR APIs (including Bulk FHIR and Subscriptions), fund CSP-backed digital identity integration, and embed interoperability requirements into both clinical and SDOH workflows. These changes would reduce friction, improve care outcomes, and make value-based care models more scalable and equitable.

VB-12. What technology standardization would preserve program-specific flexibility while promoting innovation in APM technology implementation?

To preserve flexibility across Alternative Payment Models (APMs) while promoting innovation, CMS and ONC should standardize foundational technology elements that enable consistent,

interoperable infrastructure, allowing for implementation freedom at the application and workflow layers. Key recommendations include:

1. Standardize Core Data Exchange APIs

- Mandate FHIR R4 APIs (including Patient Access, Provider Access, Payer-to-Payer, and Prior Authorization APIs) for all certified health IT systems participating in APMs.
- Require support for Bulk FHIR and FHIR Subscriptions to enable population-level risk management and event-driven care coordination.
- Encourage use of SMART on FHIR for app-level extensibility, allowing developers to tailor experiences for specific clinical or population needs.

2. Enforce Common Terminologies and Data Models

- Align all reporting and exchange requirements with standard vocabularies (e.g., SNOMED CT, LOINC, ICD-10, CPT, RxNorm) to ensure semantic interoperability across vendors and care settings.
- Mandate the adoption of USCDI v3+ for structured clinical data and expand it to include social determinants of health (SDOH), care team roles, and community resource data.

3. Embed Digital Identity Standards

- Require NIST 800-63-3 IAL2/AAL2 digital credentials via Credential Service Providers (CSPs) for patient and provider authentication to reduce fragmentation and enhance data access governance.
- Enable federated identity frameworks to allow participants to use the same digital ID across payers, providers, and apps, eliminating the need for redundant onboarding.

4. Define Modular Certification Pathways

- Create tiered certification modules under the ONC Health IT Certification Program to distinguish between basic-level interoperability and advanced functions, such as AI-driven risk scoring, digital navigation, and SDOH integration.
- This enables resource-constrained providers to meet minimum standards, while innovators can develop enhanced solutions that still interoperate seamlessly.

5. Promote TEFCA as the Default Exchange Framework

- Require all APM participants—including providers, payers, and technology vendors—to onboard to a QHIN and exchange data via TEFCA protocols.
- TEFCA's trust framework and technical infrastructure can reduce point-to-point complexity while ensuring scalability and data fidelity across regions and models.

By standardizing data transport (FHIR), semantics (terminology), identity (CSPs), and governance (TEFCA)—while maintaining flexibility in how vendors and APMs configure

applications and care models—CMS can stimulate market innovation without sacrificing interoperability, equity, or accountability.

VB-13. What improvements to existing criteria and standards would better support value-based care capabilities while reducing provider burden?

To better support value-based care (VBC) capabilities while reducing provider burden, CMS and ONC should enhance current health IT certification criteria and interoperability standards by focusing on the following key areas:

1. Require Real-Time, Actionable Data Access

- Mandate TEFCA connectivity and QHIN participation for all certified health IT vendors and VBC participants to eliminate fragmented data silos.
- Expand certification criteria to include Bulk FHIR and FHIR Subscriptions, enabling population-level insights and event-triggered care coordination (e.g., hospital discharges or emergency visits).

2. Streamline Quality Measurement Through Automation

- Replace QRDA-based reporting with automated eCQM submission via Bulk FHIR exports.
- Certify EHRs and registries to calculate and submit quality measures using real-time data feeds, rather than relying on manual chart abstraction or disconnected modules.
- Require standardized APIs for performance feedback loops, enabling providers to identify gaps and areas for improvement within their workflow.

3. Enhance Standards for Social Determinants of Health (SDOH)

- Expand the USCDI to include standardized, structured SDOH elements (e.g., housing, food insecurity, transportation, language needs) and make their collection and exchange a certification requirement.
- Align with Gravity Project standards and integrate SDOH data into risk adjustment, care planning, and quality metrics.

4. Certify Digital Tools that Minimize Documentation Burden

- Include certification pathways for AI-assisted clinical documentation, voice-to-text summarization, and contextual decision support powered by longitudinal data.
- Promote CDS Hooks and SMART on FHIR integration as baseline functionality for clinical systems.

5. Standardize Attribution and Patient Matching

- Require support for probabilistic and referential patient matching algorithms to improve care attribution in VBC programs.

- Include a federated identity infrastructure using CSP-verified credentials to simplify patient identity verification across systems and reduce redundant intake and consent processes.

6. Simplify Administrative Interoperability

- Certify APIs for prior authorization, eligibility checking, real-time benefits, and provider directory queries to reduce phone calls, faxes, and manual workarounds.
- Standardize FHIR-based intake and scheduling APIs to enable seamless integration of third-party navigation tools into practice workflows.

Improving existing certification criteria with a focus on real-time, standards-based data access, automated quality measurement, and support for SDOH and digital identity will simultaneously enable more effective VBC operations and relieve provider documentation and administrative burden. These updates will help bridge gaps in equity, accountability, and efficiency across all value-based payment models.

VB-14. How could implementing digital identity credentials improve value-based care delivery and outcomes?

Implementing NIST 800-63-3 IAL2/AAL2-compliant digital identity credentials—such as those issued by Credential Service Providers (CSPs) like Login.gov, ID.me, or CLEAR—can significantly enhance value-based care (VBC) delivery and outcomes in the following ways:

1. Enable Seamless, Patient-Centered Data Access

- **Unified patient authentication** allows individuals to access their health records across multiple provider systems, payers, and digital health tools using a single, secure login.
- This reduces information fragmentation, empowering patients to engage more consistently with their care team and digital tools, which is critical for closing care gaps and managing chronic diseases.

2. Improve Attribution and Care Coordination

- Accurate digital identity ensures precise patient matching across disparate systems, which is foundational for effective risk stratification, ACO attribution, and shared savings calculations.
- Real-time, identity-verified access enables event-driven care coordination, such as flagging hospital discharges or missed preventative screenings across care teams.

3. Enhance Consent Management and Proxy Access

- CSP-backed credentials allow patients to **delegate access to caregivers**, assign proxies, and manage their health data sharing preferences through standardized, auditable workflows.

- This facilitates informed decision-making for families managing complex care needs or multiple chronic conditions.

4. Reduce Administrative Burden and Inefficiencies

- Providers can rely on **federated identity verification** for tasks such as digital intake, remote monitoring enrollment, and cross-entity record retrieval, streamlining workflows and reducing duplicative documentation.
- These efficiencies translate into more time for patient interaction and fewer errors, which directly affect VBC quality and satisfaction metrics.

5. Build Trust and Improve Equity

- Digital identity frameworks that support multilingual, mobile-friendly, and biometric options ensure access for underserved and digitally inexperienced populations.
- Uniform identity verification helps eliminate bias from manual matching processes, enabling the equitable delivery of decision support and outreach efforts based on real-time data.

By integrating digital identity credentials across provider, payer, and patient systems, CMS can establish a trusted, scalable infrastructure that ensures patients are accurately identified, their data is shared appropriately, and care is coordinated efficiently—ultimately driving better outcomes, reduced costs, and stronger accountability in value-based care.

VB-15. How could a nationwide provider directory of FHIR endpoints help improve access to patient data and understanding of claims data sources? What key data elements would be necessary in a nationwide FHIR endpoints directory to maximize its effectiveness?

VB-15. Value of a Nationwide Provider Directory of FHIR Endpoints

A nationwide provider directory of FHIR endpoints would be a foundational asset for advancing data interoperability, especially in the context of value-based care (VBC). It would help providers, payers, and technology developers identify and access the right systems for retrieving both clinical and claims data, while enabling automated, scalable workflows across care settings.

1. Improve Access to Patient Data Across Care Teams

- A centralized FHIR endpoint directory enables automated routing of clinical queries (e.g., labs, care plans, ADTs) to the correct organization, reducing manual lookup and fax-based workflows.
- Providers participating in VBC models could more easily retrieve longitudinal patient histories from disparate EHR systems, ensuring accurate risk stratification and care gap closure.

2. Strengthen Claims Data Transparency

- By mapping FHIR endpoints to specific payer organizations, such a directory would facilitate tracing the source of claims data, thereby supporting the validation of diagnoses, services rendered, and reimbursement logic.
- This visibility would enhance trust and accuracy in risk adjustment and shared savings calculations.

3. Facilitate Real-Time Care Coordination

- With a comprehensive directory, care coordination tools can dynamically access a patient's known providers, enhancing referral tracking, handoffs, and follow-up workflows—key components of successful Accountable Care Organization (ACO) participation.

4. Enable Smart Routing for Digital Health Tools

- A public directory enables third-party applications to integrate FHIR endpoint discovery into scheduling, consent, or Clinical Decision Support (CDS) tools, thereby reducing onboarding friction and improving the time-to-value for new Value-Based Care (VBC) programs.

Key Data Elements to Include

To maximize effectiveness, a national FHIR endpoint directory should include:

- Organization name and NPI
- Endpoint URL (SMART on FHIR, Bulk FHIR, etc.)
- FHIR version supported (e.g., R4, R5)
- Endpoint type (clinical, payer, directory, scheduling, prior auth, etc.)
- Credentialing info (e.g., CSP integration, security tokens)
- Contact information for technical support
- FHIR capability statement link
- Participation in TEFCA/QHIN (yes/no)
- Supported transaction types (patient access, provider access, prior auth, etc.)
- Geographic region served
- Organizational affiliations (e.g., health system, ACO, MCO)

A comprehensive, up-to-date directory of FHIR endpoints would serve as a critical resource for national-scale interoperability, particularly for stakeholders involved in value-based arrangements. It would reduce latency, enhance transparency, and facilitate more proactive care coordination, ultimately improving outcomes and lowering costs across the healthcare ecosystem.

Thank you for the opportunity to provide my comments.

Sincerely,

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